

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



국립 한국해양대학교
NATIONAL
KOREA MARITIME & OCEAN UNIVERSITY

Made by Minho Park (Ph.D.)

<https://orcid.org/0000-0002-6900-4566>

Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas
Aftertreatment Systems

Midterm

**Ch12_Tools, Apprentice Engineer Duties, and Career
Direction**

Final Exam

Tools

The place with the most tools in the engine room is the workshop.

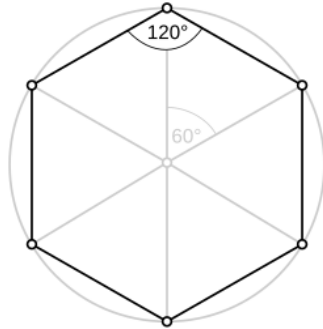
Some tools needed for work are also placed near the M/E and G/E.



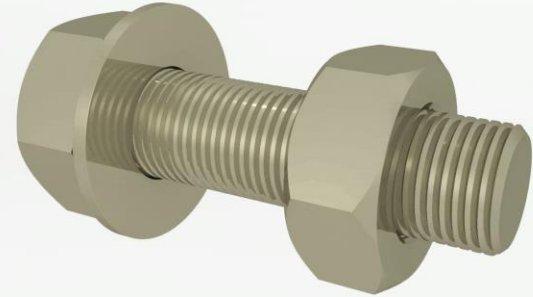
Bolt and nut

Hexagon bolt (= Hexagon head bolt, Hex bolt)

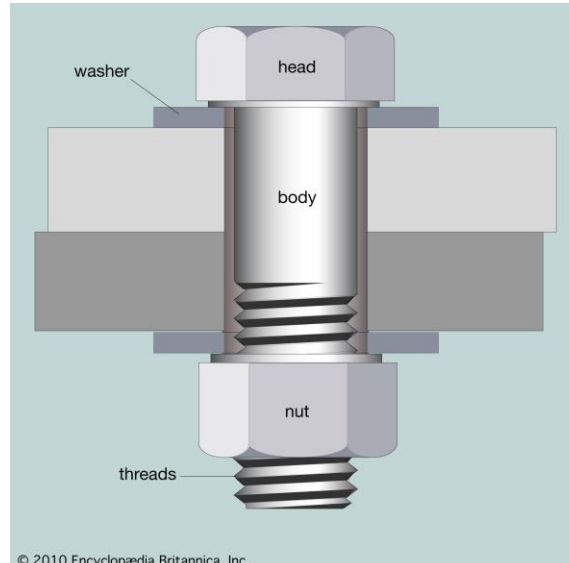
Hexagon



Bolts

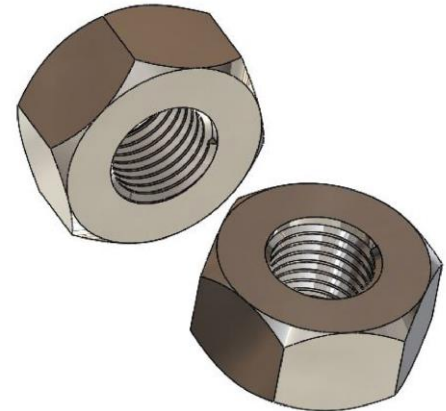


IQSdirectory.com



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Nut



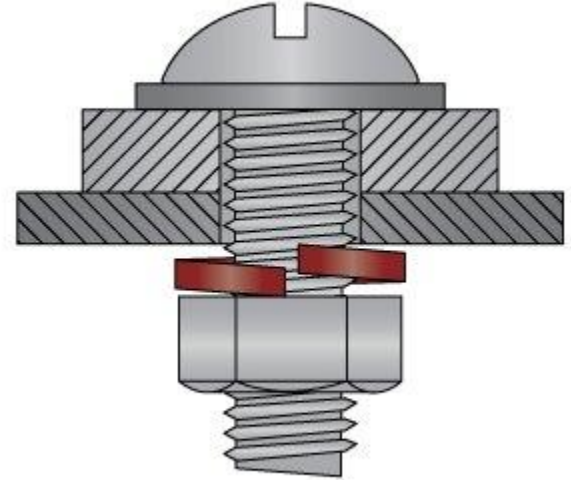
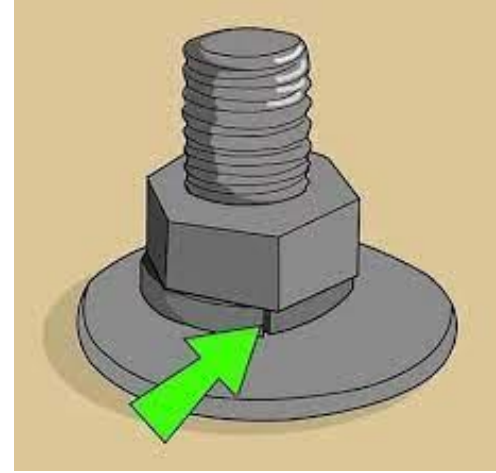
IQSdirectory.com

Washer

Plain washer (= Flat washer)



Spring washer



Circlip

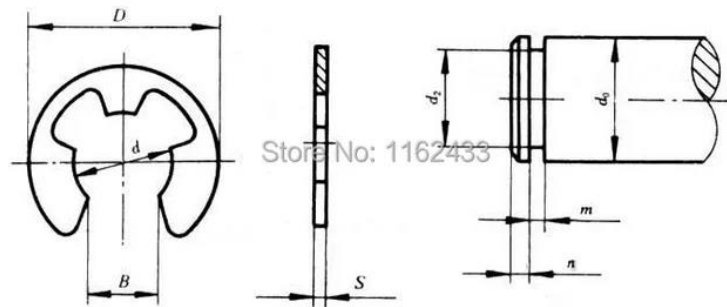
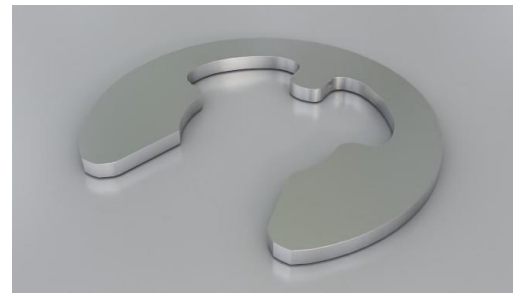
Internal circlip



External circlip



E type circlip

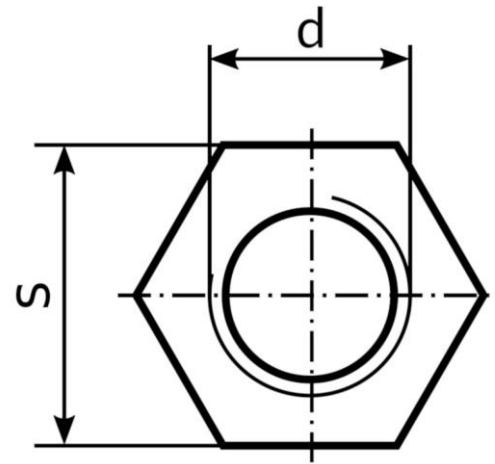
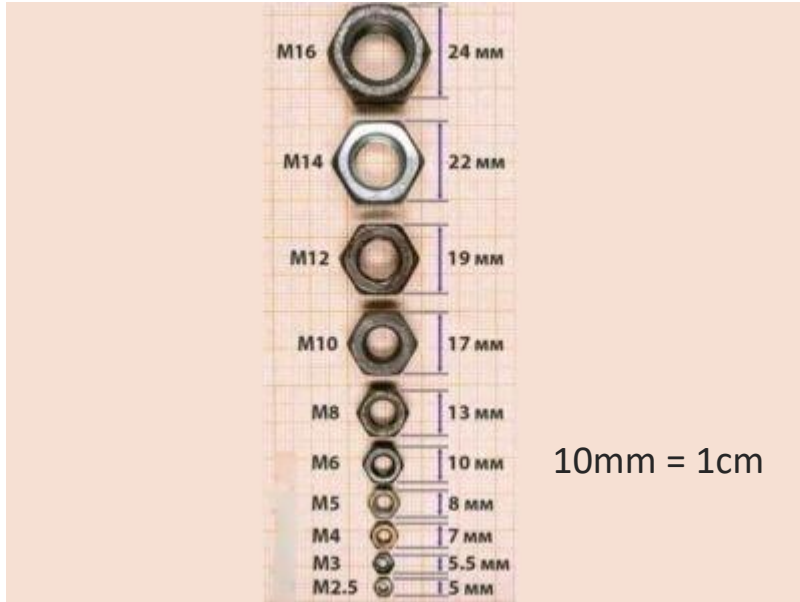


Tools

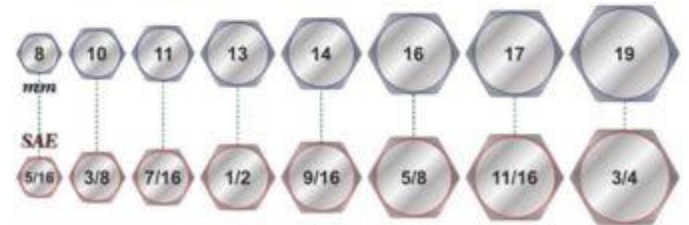
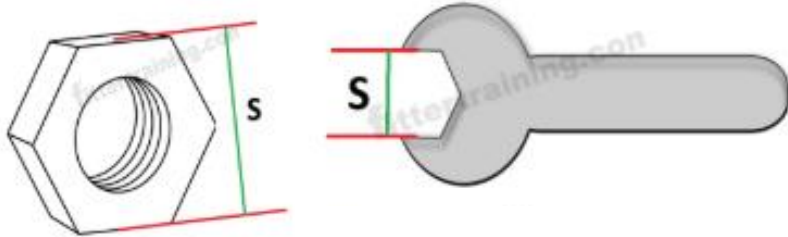
Single ended open spanner



Tools



d: nominal thread diameter
s: wrench size



Tools

Metric Spanner Size Chart for Bolts

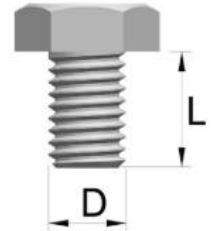
Bolt Diameter (mm)	Spanner Size (mm)
M4	7
M5	8
M6	10
M8	13
M10	17
M12	19
M14	22
M16	24
M18	27
M20	30
M22	32
M24	36
M27	41
M30	46
M33	50
M36	55
M39	60



Thread
Diameter

D

M8



Tools

Double ended open spanner



Tools

Ring spanner



Tools

Combination spanner



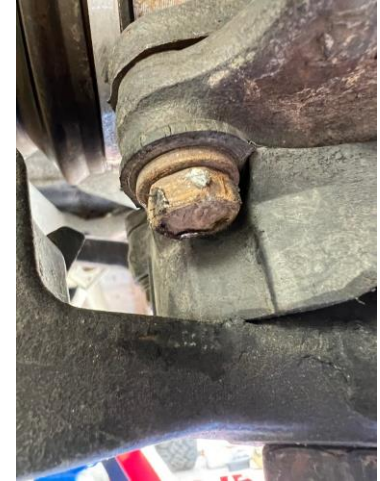
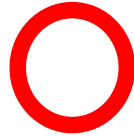
Tools



Tools



Bolt and nut



Tools

Adjustable spanner (=Monkey spanner)



Tools



Tools

Striking wrench (ring type)



Tools

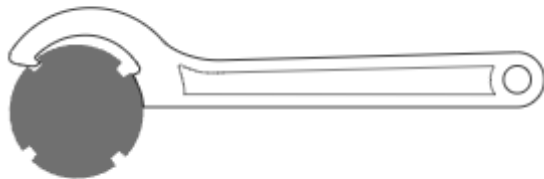
Striking wrench (open-end type)



Tools

Hook spanner

【How to use】

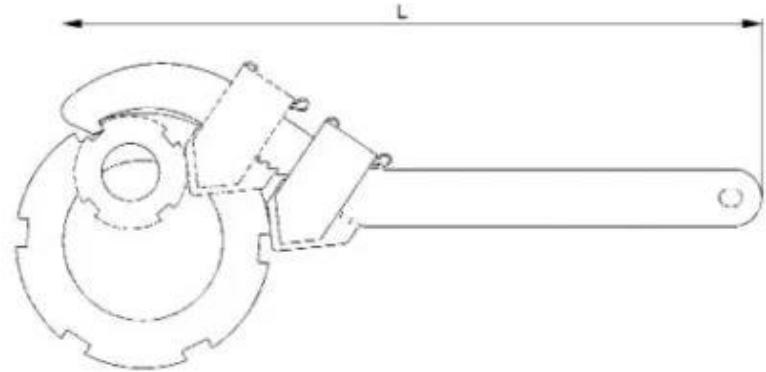


Hooks into the groove of a grooved round nut.



Tools

Adjustable hook spanner



Tools



Tools

Pipe wrench



Tools

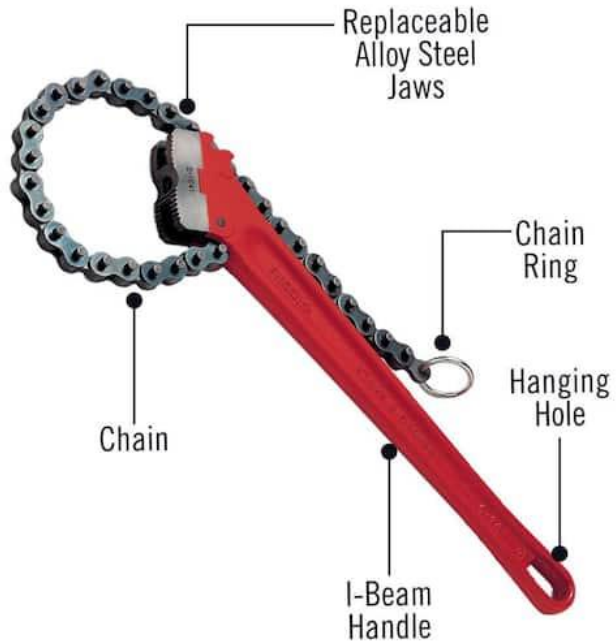


Tools



Tools

Chain pipe wrench



Tools



Tools

Socket and
Ratchet wrench



Ratchet wrench



Socket



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Tools

Working Principle of Clicker Torque Wren



Tools



Tools



Tools

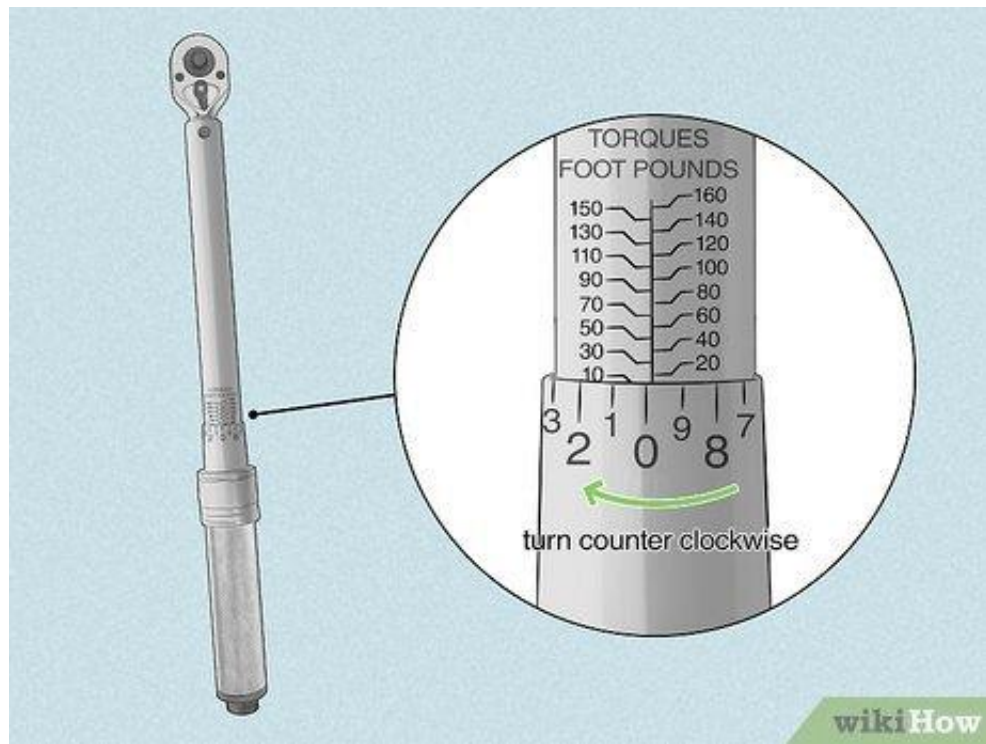


Tools

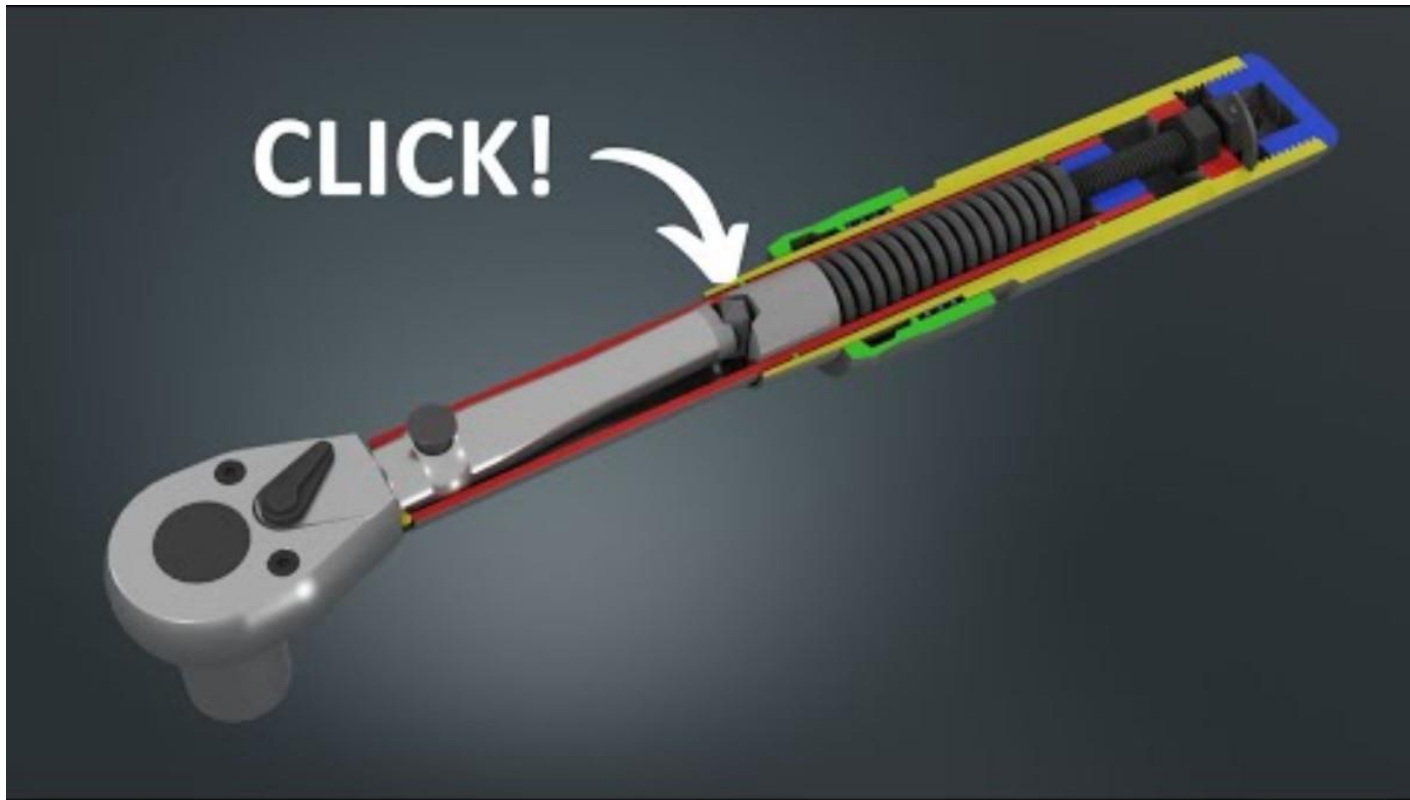


Tools

Torque wrench



Tools



Tools

T-Handle socket wrench



Tools

Ratchet spanner



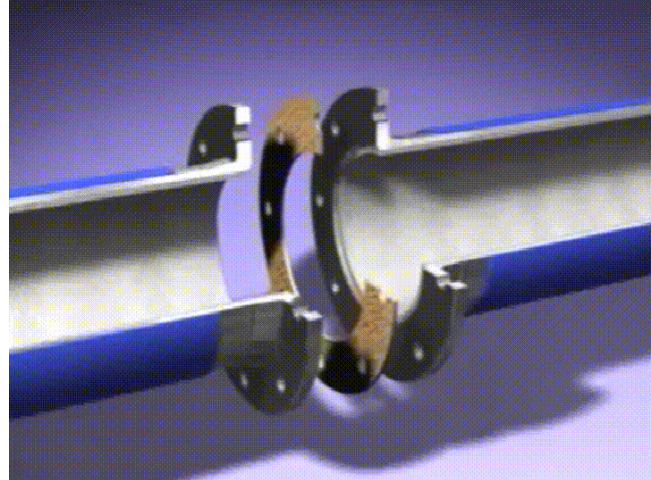
Tools

Combination ratchet spanner



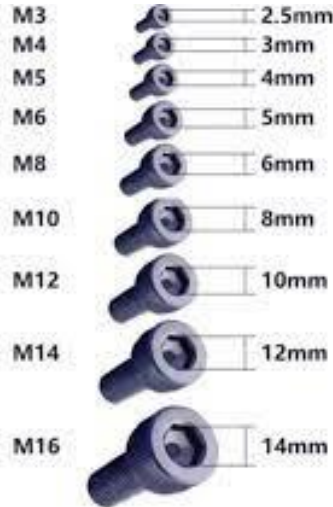
Tools

Double-ended socket ratchet wrench (spike type)



Bolt

Allen head bolt (= Allen key head bolt, Socket head bolt, Hex socket head bolt)



Tools

Allen wrench (= Allen key, Hex wrench, Hex key)



Ball-end hex wrenches



Ball End

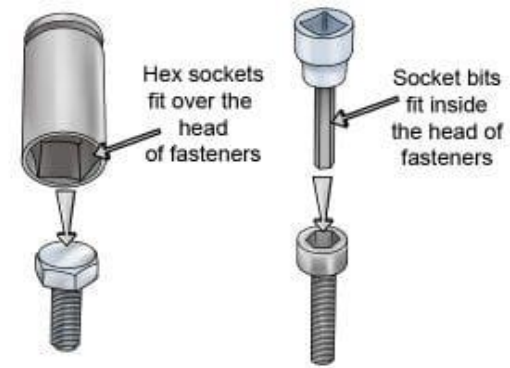
20° Ball Angle Increases Access

The ball end increases your ability to access fasteners while still maintaining sufficient strength.



Tools

Allen socket (= Allen key socket, Hex key socket)



Tools

Impact wrench (= Air impact wrench,
Pneumatic impact wrench)



Tools

Pneumatic derusting brush



Tools

Bench grinder with grinding wheel and wire brush



Never use glove for rotating machine

Mind finger cutting accident



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Tools



Tools

Hammer



Steel hammer



Copper hammer



Rubber hammer



Teflon hammer

Plastic hammer



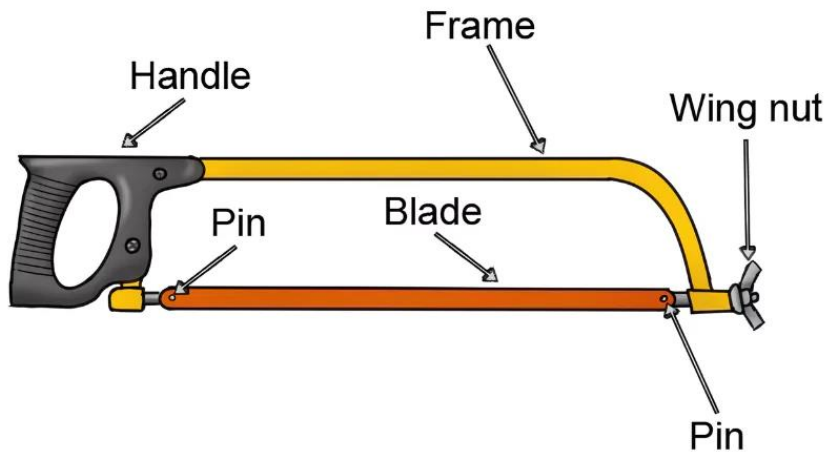
Tools

Hacksaw



What Is Hacksaw?

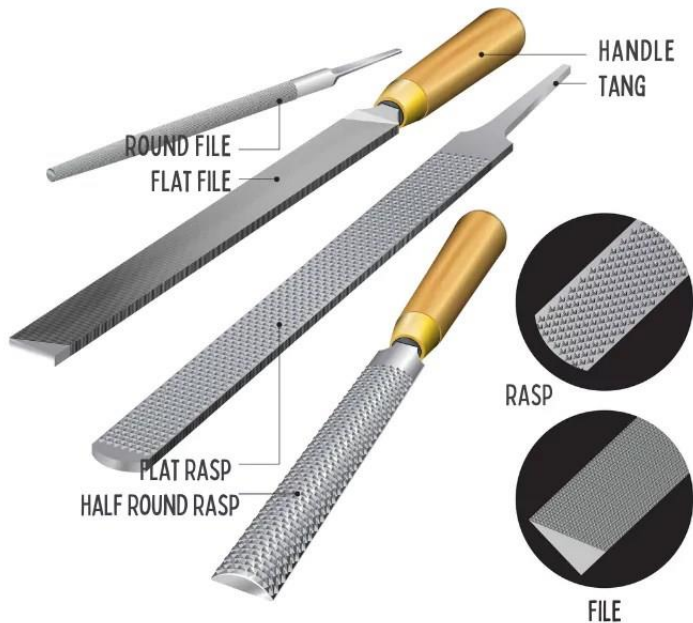
A hacksaw is a fine-toothed saw, originally and mainly made for cutting metal. The equivalent saw for cutting wood is usually called a bow saw.



Tools

File

FILE TOOL



- Half Round File - SDBC0821
- Square File - SDBA0808
- Round File - SDBD0808
- Flat File - SDBB0821
- Triangle File - SDBE0815

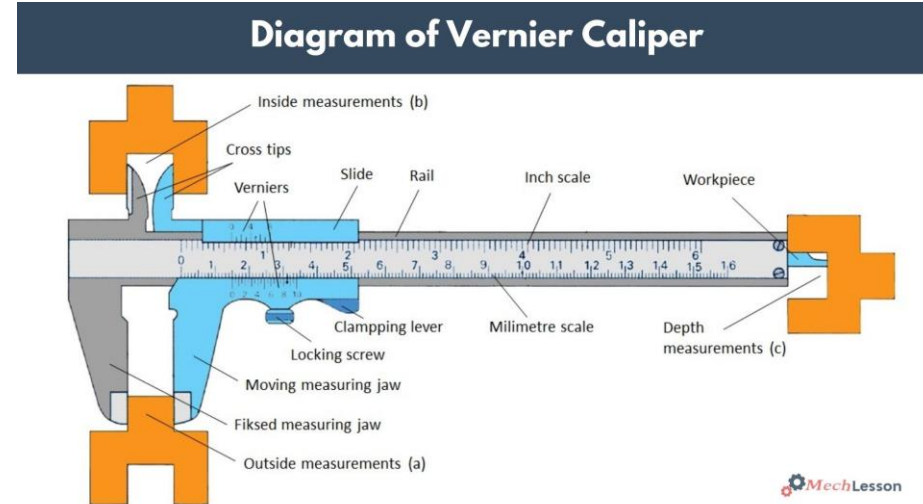
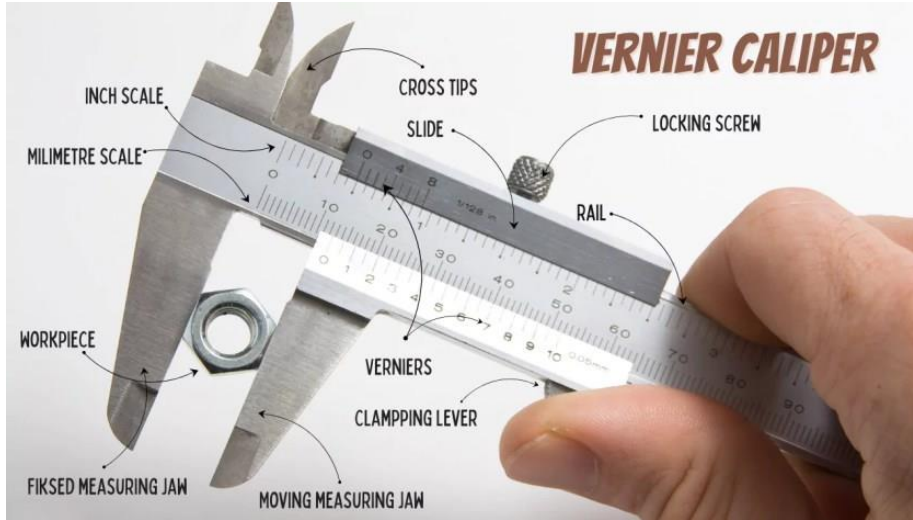
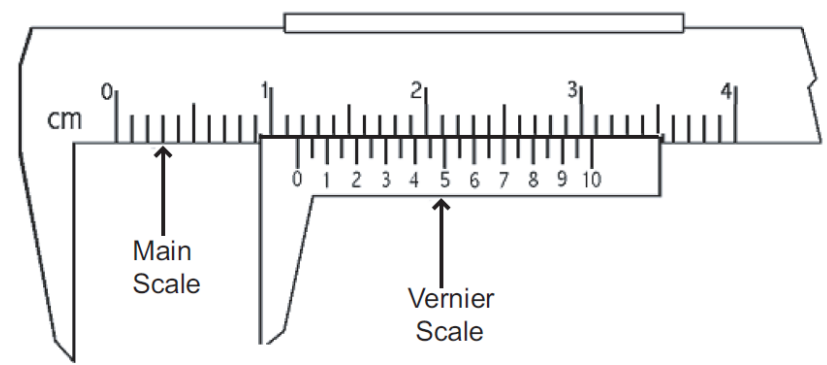
Tools

Sand paper



Tools

Vernier caliper



Tools

Vise plier (= Vise grip plier, Locking plier)



Tools

Long nose plier



Tools

Combination plier



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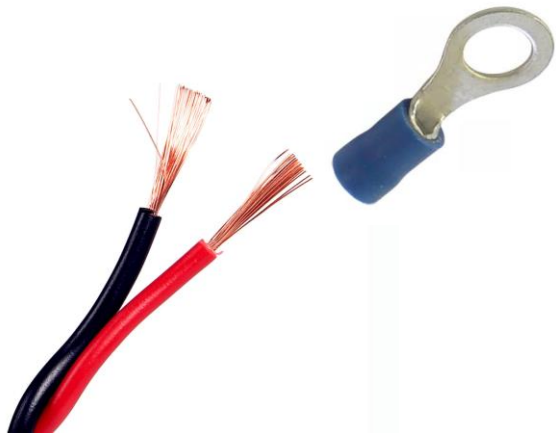
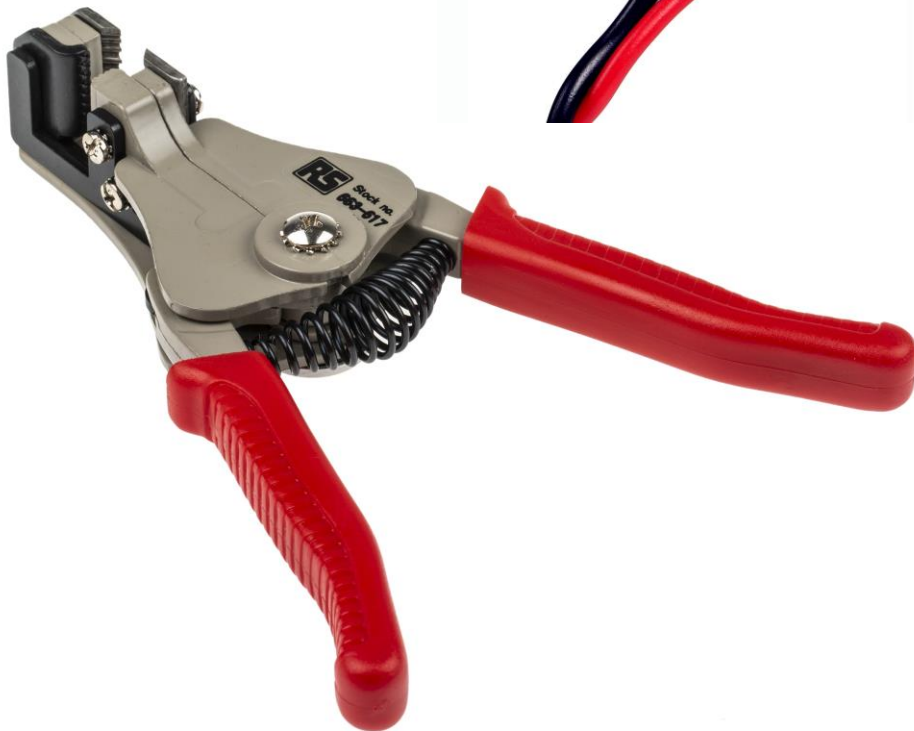
Tools

Diagonal cutting plier (= Nipper)



Tools

Electric wire stripper



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Tools



Tools

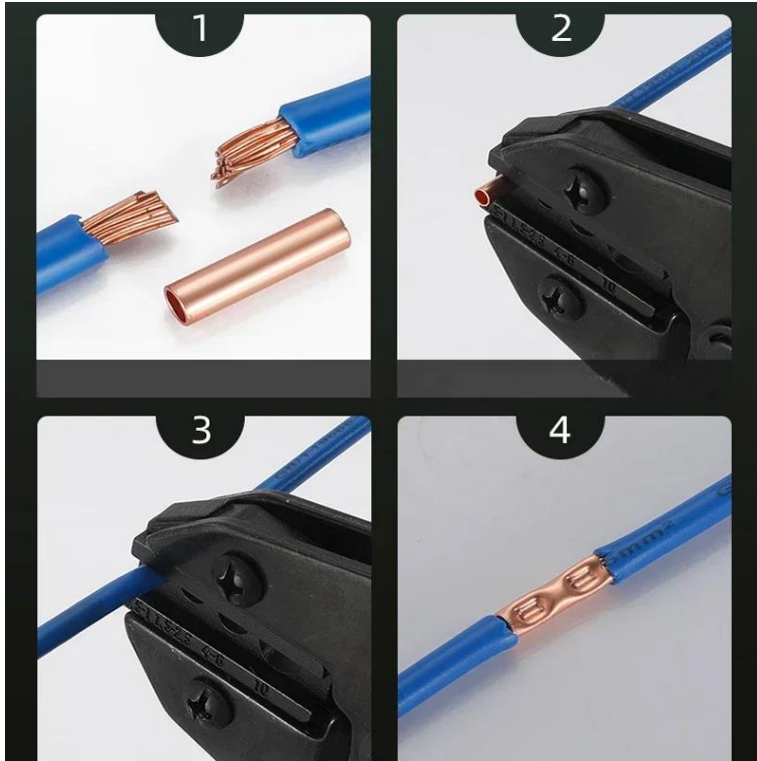
Ratchet crimping tool

(Crimping plier, Cable shoe plier)



Tools

Ratchet crimping tool (Crimping plier, Cable shoe plier)



Firmly and Simple Connection



Tools



Tools

Circlip plier



Tools



Tools

Punching tool set



Tools

<https://www.tiktok.com/@engineroominsight/video/7535494395844463879>

Tools



Tools



Tools

Hollow punch set



Tools



Tools

Digital multi tester



Tools

Digital insulation resistance tester (Digital ohm meter)



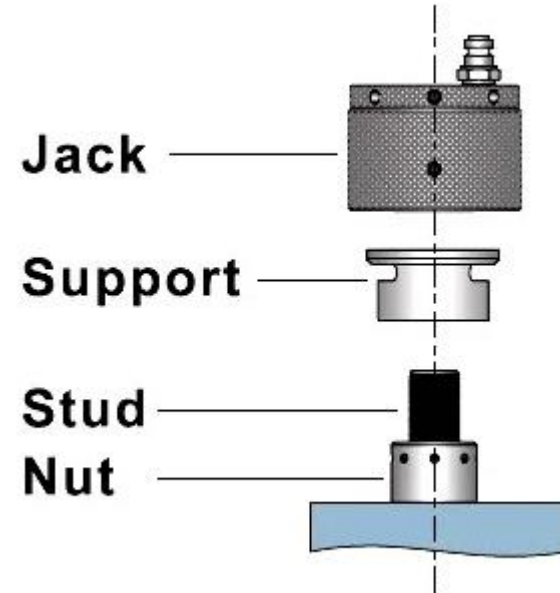
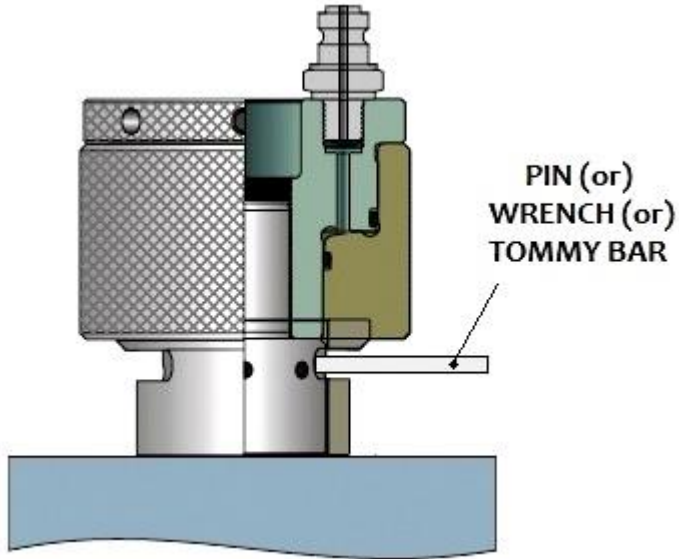
Tools

G/E cylinder head

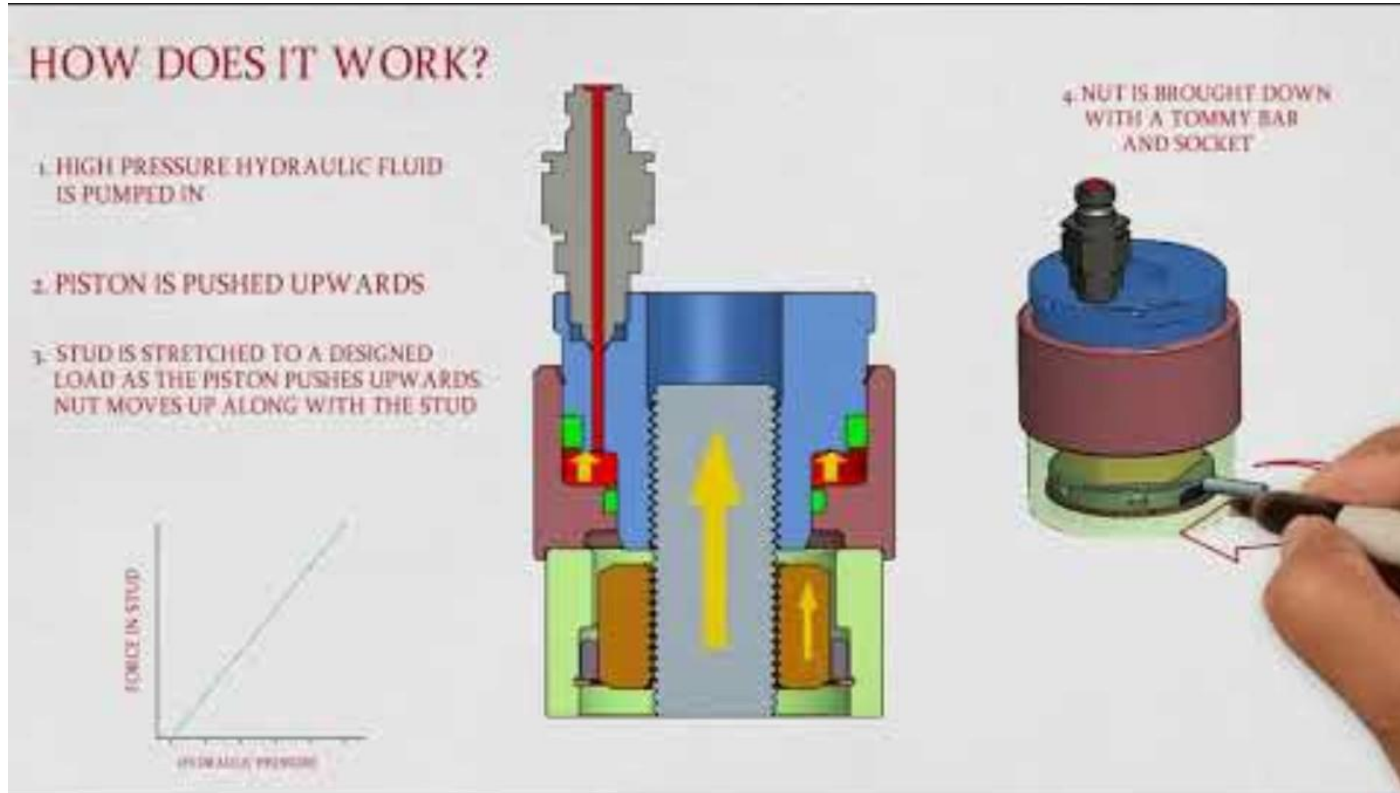


Tools

Hydraulic jack



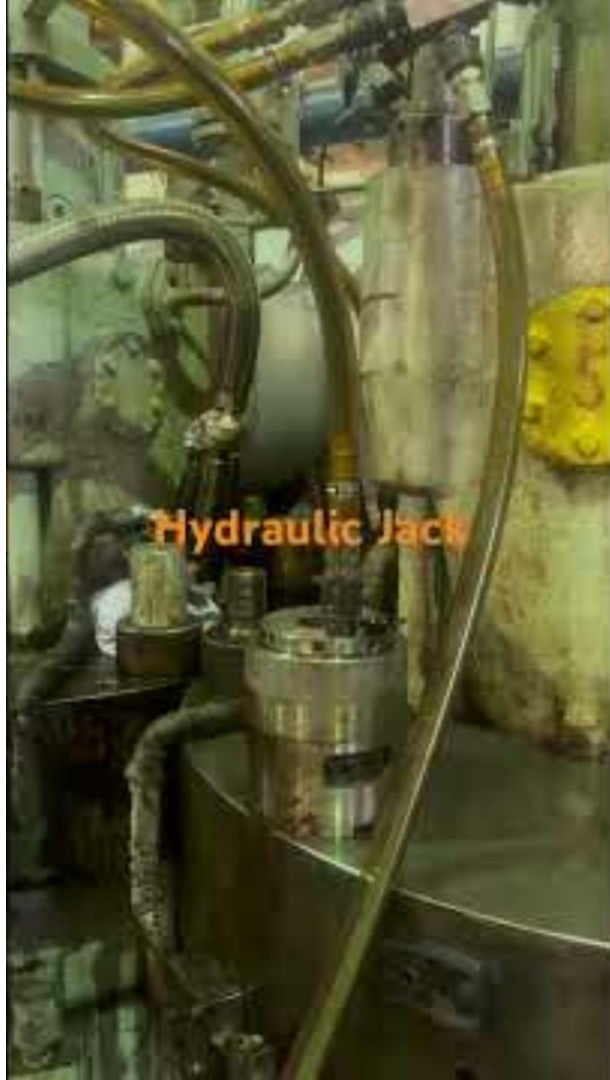
Tools



Tools



Tools



Apprentice engineer (A/E) Duties

In reality, an apprentice engineer is neither responsible for nor in charge of anything done on a ship.

However, I would like to introduce the work that an apprentice engineer would typically do on a ship. This may vary slightly from company to company and vessel to vessel, but I personally don't think it will vary much.

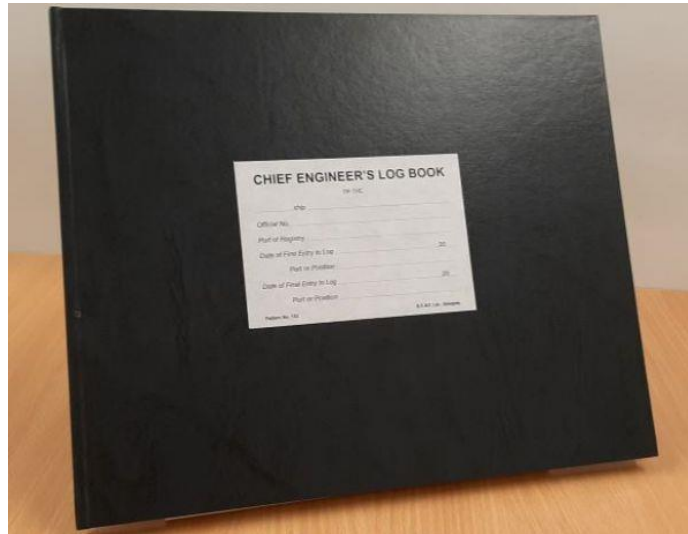
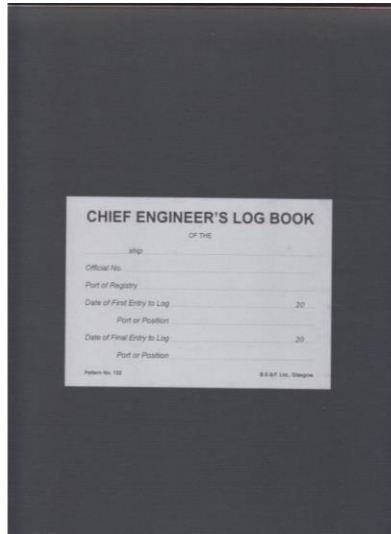
One of the most important duties of an apprentice engineer is writing a log book (= Chief engineer's log book).

The log book written at 12:00 is usually written by the third-class engineer, and at 20:00, it is written by the on-duty engineer (sailing: 3rd → 2nd → 1st, in port: 3rd → 2nd).

However, sometimes the log book may be filled out by an apprentice engineer at 12:00 and 20:00.

Apprentice engineer (A/E) Duties

A log book looks like this, and while each ship and company has its own unique log book, its basic purpose remains the same.



A log book looks like this, and while each ship and company has its own unique log book, its basic purpose remains the same.

Engineers' Log M.V. _____ Voyage No. _____

[illegible][illegible]

MAINEENIGNES	AUXILIARIES	BOILERS

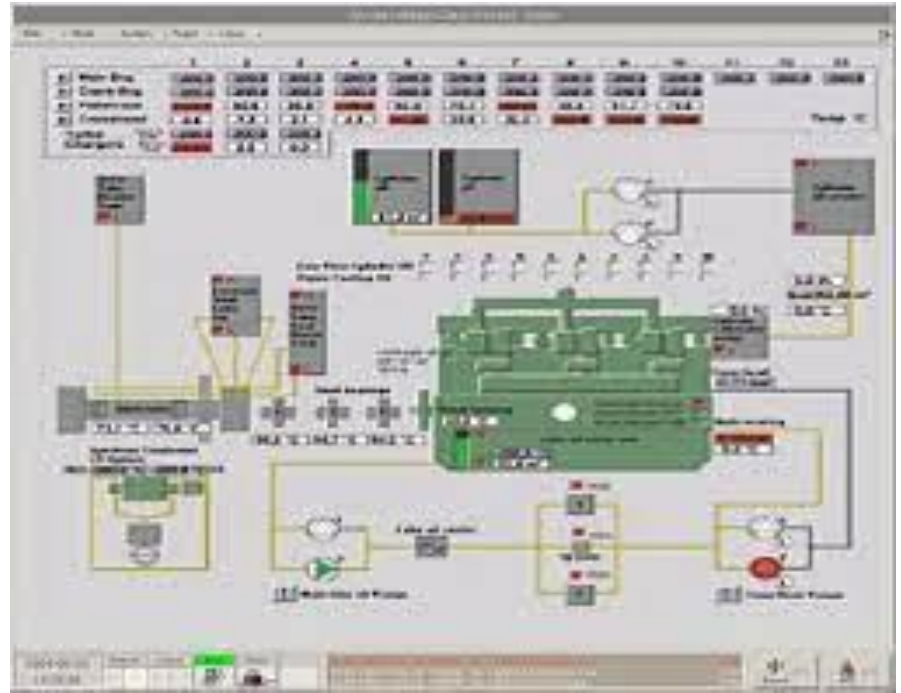
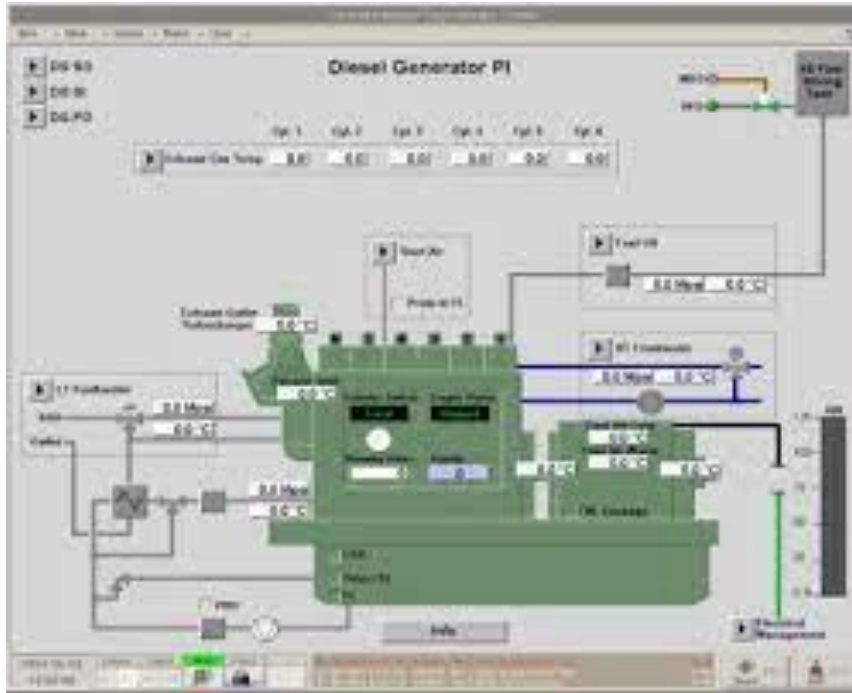
<i>from</i>	<i>to</i>	<i>Date</i>

[illegible]

STEAM BOILER IN USE	Hours	Min.	CONSUMPTION OF FUEL OIL BY MAIN ENGINES				Tonnes	EXTENDED REMARKS
STEAM PRESSURE			CONSUMPTION OF FUEL OIL BY AUXILIARY MACHINERY				Tonnes	
CONSUMPTION OF FUEL OIL BY BOILER	Tonnes		TOTAL				Tonnes	
DESCRIPTION OF FUEL OIL IN USE			FUEL OIL PER LAST ACCOUNT					
VISCOSITY OF FUEL OIL IN USE AT 60° C	CGS		FUEL OIL RECEIVED AT					
FLASHPOINT OF FUEL OIL IN USE	°C		FUEL OIL USED NOON TO NOON					
SPECIFIC GRAVITY OF FUEL OIL IN USE			FUEL OIL REMAINING					
HOURS UNDER WAY	Hours	Min.	OILS REMAINING ON BOARD					
HOURS AT FULL SPEED	Hours	Min.						
DISTANCE RUN BY OBSERVATION								
SPEED		Knots	Lubricating Oil	Cylinder Oil	Compressor Oil	HFO/LSHFO	HFO/LSHFO	
AVERAGE REVOLUTIONS PER MINUTE	Rev.	Sec.	TOTAL					
DISTANCE BY ENGINES			EXPENDED					
SLIP OF PROPELLER			REMAINING					
TOTAL DISSOLVED SOLIDS			Engineer on Watch Chief Engineer					
ALKALINE								
ON SCAVENGE								

Apprentice engineer (A/E) Duties

Alarm monitoring system (AMS) in the engine control room (ECR)



Apprentice engineer (A/E) Duties

Thermometers and pressure gauges attached to the machinery



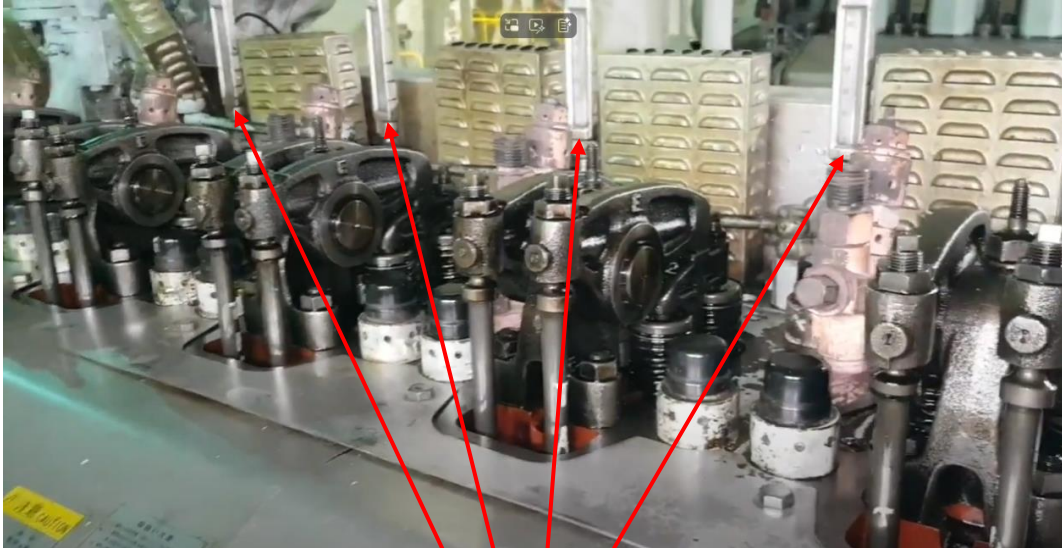
Pressure gauges



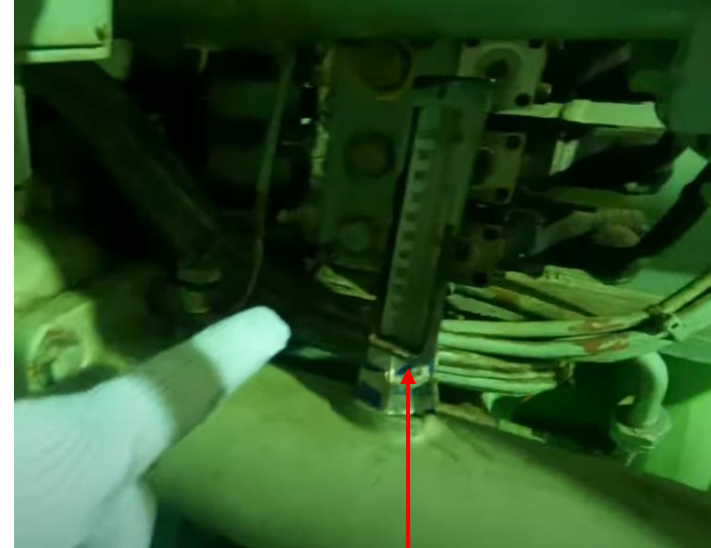
Digital value based on sensor, not gauge value

Apprentice engineer (A/E) Duties

Thermometers and pressure gauges attached to the machinery



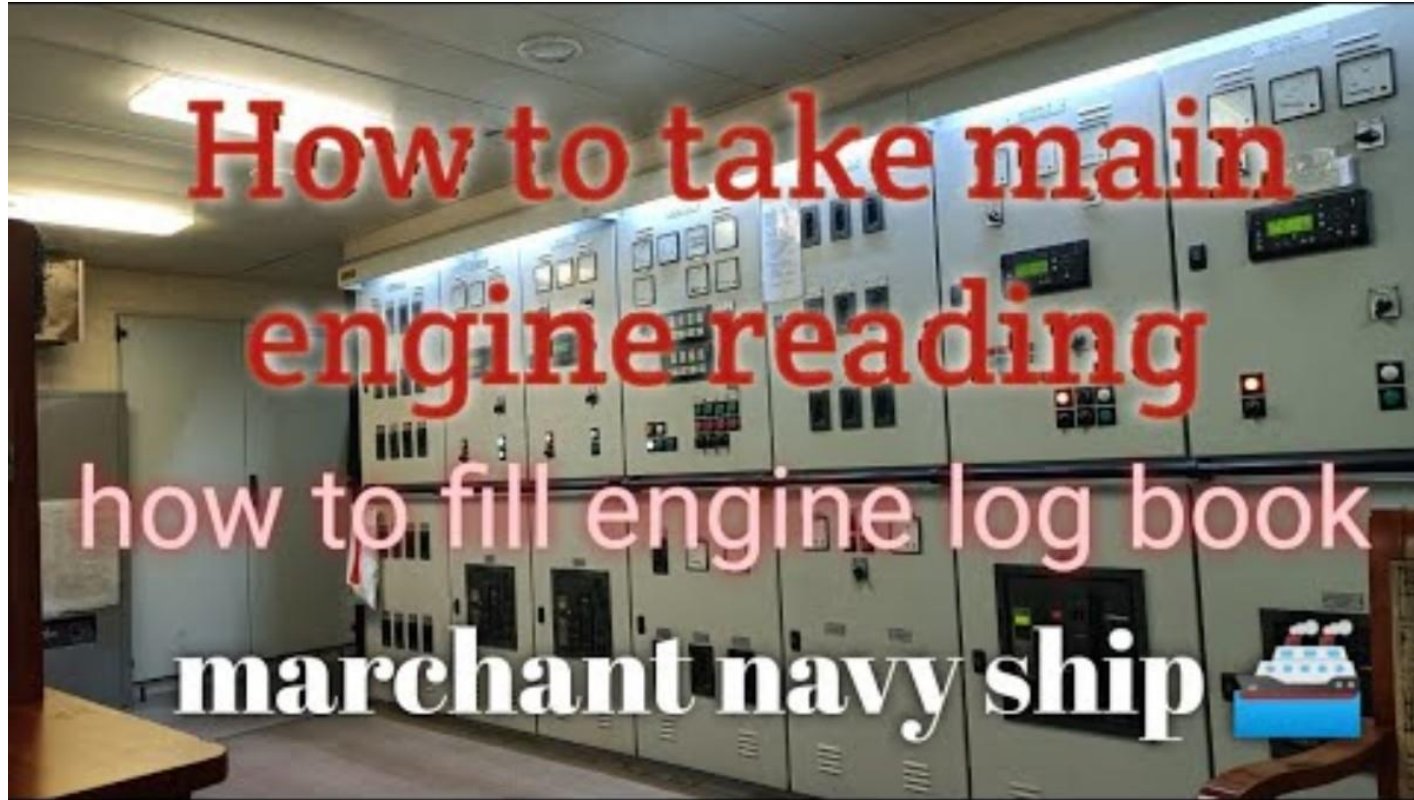
Thermometers



Thermometers



Apprentice engineer (A/E) Duties



Apprentice engineer (A/E) Duties

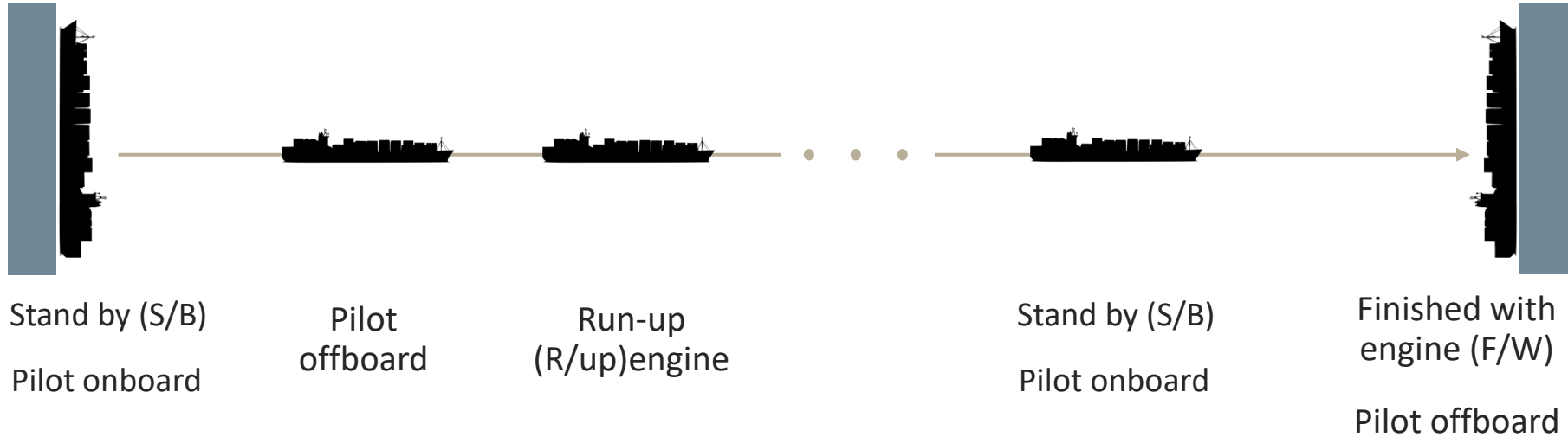


Apprentice engineer (A/E) Duties



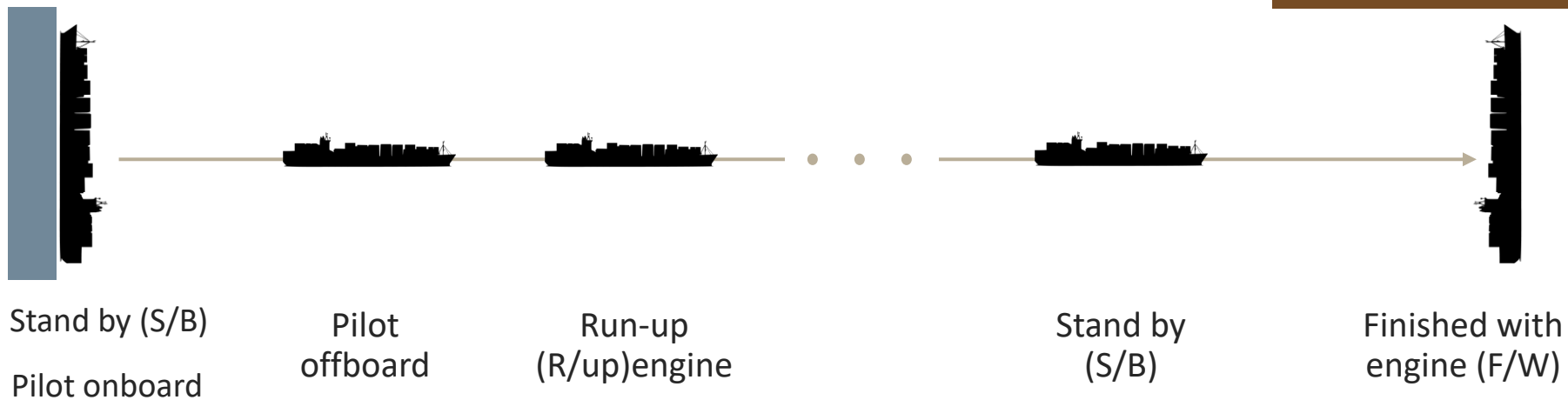
Apprentice engineer (A/E) Duties

Port to Port



Apprentice engineer (A/E) Duties

Port to Anchorage



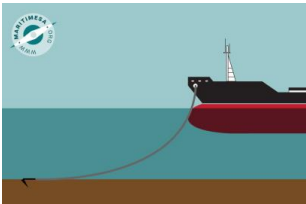
Apprentice engineer (A/E) Duties

Anchorage to Port



Stand by (S/B)

Pilot onboard



Finished with
engine (F/W)

Pilot offboard

Apprentice engineer (A/E) Duties

In the case of very old ships, a duty system was implemented and the duty system was maintained as follows.

1st engineer: 04:00~08:00 / 16:00~20:00

2nd engineer: 00:00~04:00 / 12:00~16:00

3rd engineer: 08:00~12:00 / 20:00~24:00

However, as ships became automated, the watch system was stopped and the unattended machinery space (UMS) system was started. You can assume that all the ships you will be boarding are using the UMS system.

UMS is also called as unattended machinery area (UMA).

Apprentice engineer (A/E) Duties

The UMS system operates with 1st, 2nd, and 3rd engineers working together from 08:00 to 17:00, and then a duty system (3rd → 2nd → 1st) is implemented during the evening and early morning hours (17:00 to 08:00). (Times may vary slightly depending on the company and vessel.)

When you use the UMS system, you will be required to write a UMS checklist at 16:00 to check and record the status of most machines in the engine room.

The UMS checklist, like the log book, reads and records the values in the AMS and on-site (gauges).

Apprentice engineer (A/E) Duties

HANJIN SHIPPING									
ENGINE DEPARTMENT U.M.A. CHECK LIST									
(기타식 운항 목적지 1600시에 승선로 출하 기지)									
M/V HANJIN BREMERHAVEN									
VOY. NO. 003888									
NO	CHECK POINT	DATE	DATE	DATE	DATE	DATE	DATE	DATE	DATE
		06.22	06.23	06.24	06.18	06.19	06.20	06.21	06.22
1st	AIR COND. / FAN ROOM	A/C CENT. FAN AMP(N:34) / COND.							
		AIR IN / OUT TEMP.							
		R-404A OUT TEMP NO.1/NO.2							
		LOGPCONDITION/DEGREE OF SUPERHEAT							
2nd	REF. CHAMBER TEMP	MEAT ROOM(N:-20)							
		FISH ROOM(N:-20)							
3rd	FUNEL. EXH. GAS- CONDITION	VEGETABLE ROOM(N:3)							
		R/E							
		2/E							
4th	E/R FAN ROOM CONDITION (PORT / STBD SIDE)								
5th	BOILER & INC. EXH. PIPE DRAIN TANK CONDITION								
6th	INCINERATOR D.O TK / W.O TANK LEVEL								
7th	EXH. GAS ECONOMIZER METER								
8th	INCINERATOR CONTROL PANEL CONDITION								
9th	AUX. BOILER DRUM WATER LEVEL(NL)								
10th	AUX. BOILER	DRUM STEAM PRESS. (5.0-7.0)							
		P.O TEMP(N:116) / PANEL COND.							
		CHEM. DOSING TANK LEVEL(1/6T)							
11th	H.T.C.F.W. EXPANSION TK. LEVEL (1.2T)								
12th	L.T.C.F.W. EXPANSION TK. LEVEL (0.8T)								
13th	WORK SHOP & STORE ROOM ARRANGEMENT								
14th	WORK SHOP COND. / TEMP								
15th	UNIT COOLER PRESS. SUC. / DISCH. (N:0.5/1.8)								
16th	CYL. OIL MEASURING TANK LEVEL	IN RUN / MOTOR AMPARE							
		PROV. REF. COMP. UNIT							
		LEVEL							
		COMP. L.O.(N:1/2)							
		CONDENSER R-404A(N:1/3)							
		SUC./DISCH. (N:1.0/16.6)							
17th	No. 1 AIR COND. COMP. UNIT	LEVEL							
		COMP. L.O.(N:1/2)							
		CONDENSER R-404A(N:1/3)							
		SUC./DISCH. (N:0.18/0.18)							
		LO. (N:4.0)							
		MOTOR AMPARE							
18th	No. 2 AIR COND. COMP. UNIT	LEVEL							
		COMP. L.O.(N:1/2)							
		CONDENSER R-404A(N:1/3)							
		SUC./DISCH. (N:0.18/0.18)							
		LO. (N:4.0)							
		MOTOR AMPARE							
19th	M/E EXH. PIPE DRAIN TANK CONDITION								
20th	MINERALIZER / STERILIZER CONDITION								
21th	HOT W. CIRC. PUMP CONDITION / DIS. PRESS.								
22th	CALORIFIER CONDITION / TEMP(TO J)								
		주요자 직적 & SIGN							
	CHIEF ENGINEER'S SIGNATURE								

NO	CHECK POINT	DATE	DATE	DATE	DATE	DATE	DATE	DATE	DATE
		06.16	06.16	06.17	06.18	06.19	06.20	06.21	06.22
	EXH. GAS / J.C.F.W. TEMP								
	400 ↓ / 90 ± 3								
		NO. 1st							
		NO. 2nd							
		NO. 3rd							
		NO. 4th							
		NO. 5th							
		NO. 6th							
		NO. 7th							
		NO. 8th							
		NO. 9th							
		NO. 10th							
		NO. 11th							
		NO. 12th							
	CYL. COVER F.O. L.O. AIR LEAKAGE								
	RAIL UNIT LEAKAGE CONDITION (PID/ART)								
	M/E TOP SPRING PRESS. (N:50) (PID/ART)								
	STBD								
	SPRING AIR DRAIN TRAP (PID/ART)								
	AIR FILTER METER (70L)								
	NO. 1 T/C								
	EXH. GAS IN/OUT T. (N:430/340)								
	L.O. TEMP. / PRESS. (76 ↓ / 0.9T)								
	AIR FILTER METER (70L)								
	NO. 2 T/C								
	EXH. GAS IN/OUT T. (N:430/340)								
	L.O. TEMP. / PRESS. (76 ↓ / 0.9T)								
	AIR FILTER METER (70L)								
	NO. 3 T/C								
	EXH. GAS IN/OUT T. (N:430/340)								
	L.O. TEMP. / PRESS. (76 ↓ / 0.9T)								
	C.O. LUB. WORKING CONDITION								
	SERVO OIL AUTO FILTER COUNT / ΔP COND.								
	F.O. IN / OUT PRESSURE								
	SUPPLY / PULP. PUMP LEAK / COND								
	UNIT / SERVO OIL PUMP LEAK / COND								
	CONTROL OIL PUMP LEAK / COND								
	AIR DIFF. PRESS. (200 ↓)								
	NO. 1 AIR COOLER								
	TEMP. AIR IN/OUT (N:160/46)								
	TEMP. C.W. IN/OUT (76 ↓)								
	NO. 2 AIR COOLER								
	AIR DIFF. PRESS. (200 ↓)								
	TEMP. AIR IN/OUT (N:160/46)								
	TEMP. C.W. IN/OUT (76 ↓)								
	NO. 3 AIR COOLER								
	AIR DIFF. PRESS. (200 ↓)								
	TEMP. AIR IN/OUT (N:160/46)								
	TEMP. C.W. IN/OUT (76 ↓)								
	SPRING AIR/CONT. AIR/ST-BY AIR PRESS. (N:6.8)								
24th	WATER MIST FIRE PUMP UNIT CONDITION								
25th	BOILER FEED WATER SALINITY								
26th	INSPECTION CHAMBER CONDITION								
27th	CASCADE TANK LEVEL / TEMP. (2.0T / N:40 C)								
28th	DRAIN COOLER	TEMP. DRAIN IN / OUT							
		TEMP. C.W. IN / OUT							
29th	VACUUMATOR START/STOP PRESS. / TIME								

NO	CHECK POINT	DATE	DATE	DATE	DATE	DATE	DATE	DATE	DATE
		06.16	06.16	06.17	06.18	06.19	06.20	06.21	06.22
30th	BOILER FEED WATER TANK LEVEL								
31th	MAIN L.O. COOLER	NO. 1 L.O. TEMP. IN / OUT (N:45)							
		NO. 2 L.O. TEMP. IN / OUT (N:45)							
32th	NO. 2 CENTRAL F.W. COOLER	S.W. TEMP. IN / OUT							
		F.W. TEMP. IN / OUT (N:36)							
33th	NO. 2 CENTRAL F.W. COOLER	S.W. TEMP. IN / OUT							
		F.W. TEMP. IN / OUT (N:36)							
34th	CYL. OIL STORAGE TANK LEVEL								
35th	F.W. OEN.	SHELL VACUUM / TEMP (720/60)							
		SALINITY ALARM SET/ACTUAL (10/)	10	10	10	10	10	10	10
		DISTILLATE PUMP PRESS							
36th	M/E J.C.F.W. COOLER JACKET F.W. IN / OUT								
37th	M/E J.C.F.W. PUMP	IN RUN							
		PRESS. IN / OUT (N:1.3/4.3)							
		IN USE							
38th	CENTRAL CLO. F.W. PUMP	PRESS. IN / OUT (N:1.1/3.8)							
		PRESS. IN / OUT (N:1.1/3.8)							
39th	M.W.D. SETT. & SERV. TANK DRAINING								
40th	CONTROL OIL PUMP IN RUN / AMP(N:28)								
41th	BILOGE FIRE & G/S PUMP CONDITION								
42th	BALLAST PUMP CONDITION								
		IN USE							
43th	MAIN CLR. S.W. PUMP	PRESS. IN / OUT (N:0.6/2.6)							
		PRESS. IN / OUT (N:0.6/2.6)							
44th	F.W. GENERATOR	EJECTOR PUMP PRESS. OUT (N:4.2) / AMP							
		CU AMP NO. 1 / NO. 2 / NO. 3							
45th	M.S.P.S.	AL AMP NO. 1 / NO. 2							
		IN USE / FLOW RATE (16T)							
46th	AXIAL DETURNER	DRIVING END PRESS. (4.0-5.0)							
		PRESS. END PRESS. (4.0-5.0)							
		OIL PRESS. FOR DAMPER (4.0-5.0)							
47th	AIR COOLER DRAIN	NO. 2 AIR CLR. DRAIN. WATER / OIL							
		NO. 2 AIR CLR. DRAIN. WATER / OIL							
		NO. 1 AIR CLR. DRAIN. WATER / OIL							
48th	AIR COOLER WATER DRAIN AIR VENT COND.								
49th	POTABLE W. HYD. UNIT COND / PRESSURE (N:6.0)								
50th	F.W. HYD. UNIT COND / PRESSURE (N:6.0)								
51th	BLR FEED	IN RUN							
		WATER PUMP	PRESS. IN / OUT (N:0.5/14.6)						
52th	M/E L.O. SUMP TANK LEVEL								
53th	SHART GROUNDING SYSTEM (MV) (60 ↓)								
54th	BILOGE PUMP & BILOGE TRANS PUMP CONDITION								
55th	M/E THRUST BEARING L.O. TEMP (65 ↓)								
56th	M/E RESIDUE OIL TRAP CONDITION								
57th	OILY BILOGE PUMP CONDITION								
58th	OILY WATER SEPARATOR CONDITION								
59th	E/R BILOGE WELL CONDITION/TRANSFER								
60th	E/R OTHER TANK CONDITION/TRANSFER								



Apprentice engineer (A/E) Duties

Calculation of fuel oil consumption for M/E, G/E, and BLR

A metric ton, also known as a tonne, is a unit of mass equal to 1,000 kilograms.

It is the standard unit of mass in the International System of Units (SI) and is widely used in international trade, shipping, and many countries around the world.



Apprentice engineer (A/E) Duties

Calculation of fuel oil consumption for M/E, G/E, and BLR

HOW TO CALCULATE THE CONSUMPTION OF FUEL OIL?									
GIVEN	SPECIFIC GRAVITY OF FUEL AT 15 *C				0.9850		COEF FACTOR		0.00067
	F.O SERVICE TANK TEMPERATURE *C				90				0.00065
	FLOWMETER READING 1 DEC 2019 @ 1200H LTRS				1830320		T1		15
	FLOWMETER READING 2 DEC 2019 @ 1200H LTRS				1849650				
CONS IN M3 = $\frac{\text{FM PRES} - \text{FM PREV}}{1000} = \frac{1849650 - 1830320}{1000} = \frac{19330}{1000} = 19.33$									
CORR. S.G = SG @ 15 - T2 - T1 X COEF FACTOR									
= 0.9850 90 - 15 X 0.00067									
= 0.9850 - 75 X 0.00067									
= 0.9850 - 0.05									
CORR. S.G = 0.9348									
You don't need to calculate S.G.									
2nd engineer will let you know.									
CONS IN MT = CORR. S.G X CONS IN M3									
= 0.9348 X 19.33									
CONS IN MT = 18.07									
This value continuously change after getting new fuel oil.									

You don't need to calculate S.G.

2nd engineer will let you know.

This value continuously change after getting new fuel oil.

Apprentice engineer (A/E) Duties

1,000kg = 1 ton

Calculation of fuel oil consumption for M/E, G/E, and BLR

Density of water ($\rho_{\text{water}} = 1,000 \text{ kg/m}^3$ at 15°C)

Specific gravity (SG) measures a substance's density relative to the density of a reference substance, usually water. (SG has no units.)

Example: $\text{SG} = 0.9850 \rightarrow \text{Density } \rho = 0.9850 \times 1,000 = 985 \text{ kg/m}^3 \rightarrow 985 / 1000 = 0.985 \text{ t/m}^3$

Therefore, fuel oil consumption formula is as follows.

Example: $0.985 \times 1,000 \div 1,000 \times (\text{fuel oil consumption in m}^3) = \text{fuel oil consumption in t (ton)}$

Apprentice engineer (A/E) Duties

Air reservoir tank drain

No. 1 air reservoir tank

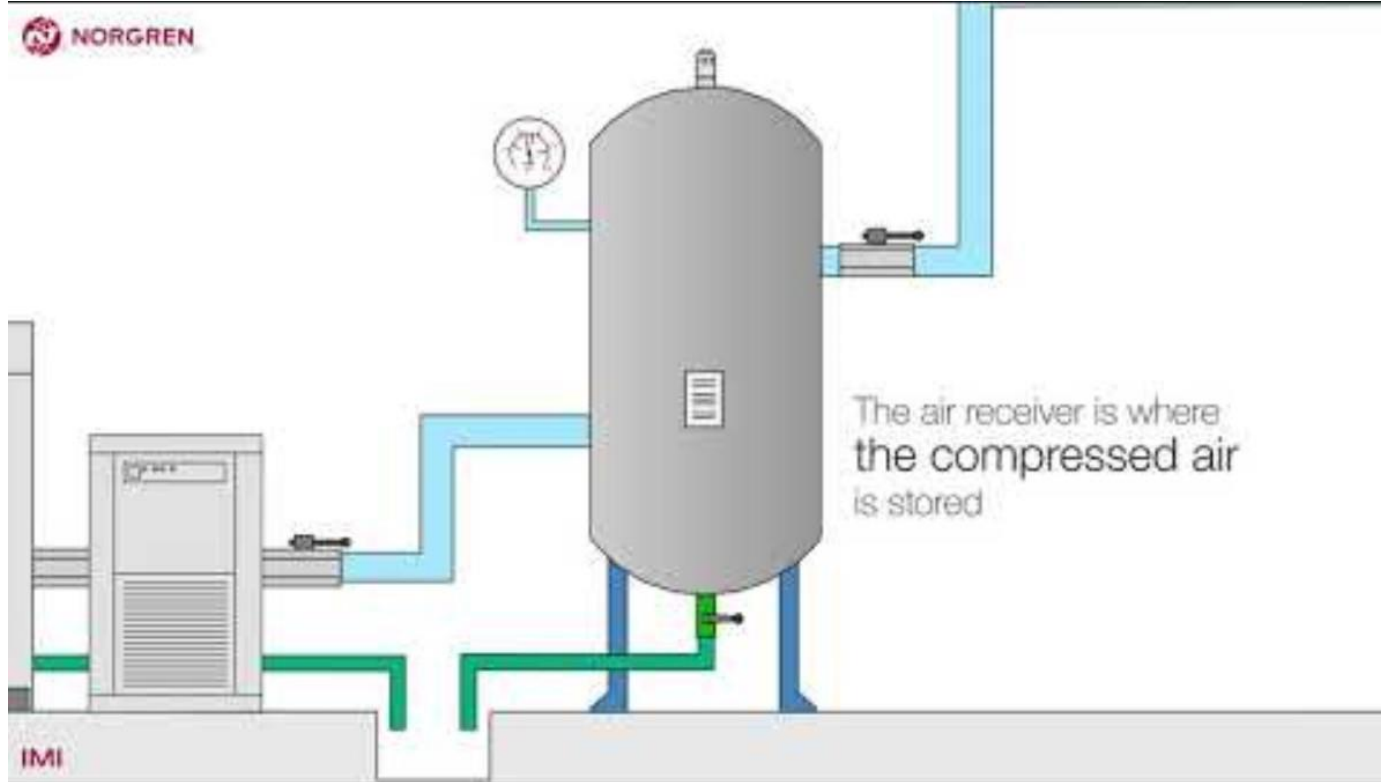
No. 2 air reservoir tank



Water drain valves

Clockwise: Valve close
Counterclockwise: Valve open

Apprentice engineer (A/E) Duties



Apprentice engineer (A/E) Duties

Fuel oil tank drain (F.O. tanks, HFO sett tank, HFO serv tank)



Apprentice engineer (A/E) Duties

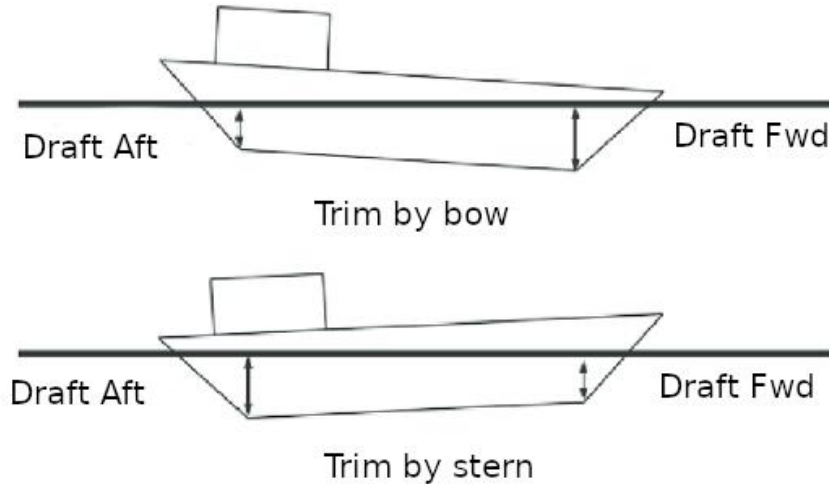
Filling of fresh water to cooling water expansion tank



GENERATOR ENGINE, NITROGEN GENERATOR THE
WATER IN THE TANK NEEDS TO BE KEPT AT THE
REQUIRED LEVEL AT ALL TIMES TO ENSURE THAT
SUFFICIENT WATER IS SUPPLIED TO THE ENGINE.

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



marine insight

SHIP TRIM

Gravitational and Buoyancy Force at Equilibrium

Even Keel

Weight added Aft causes ship to Trim by Stern

Trim by Stern

Weight added Fwd causes ship to Trim by Head

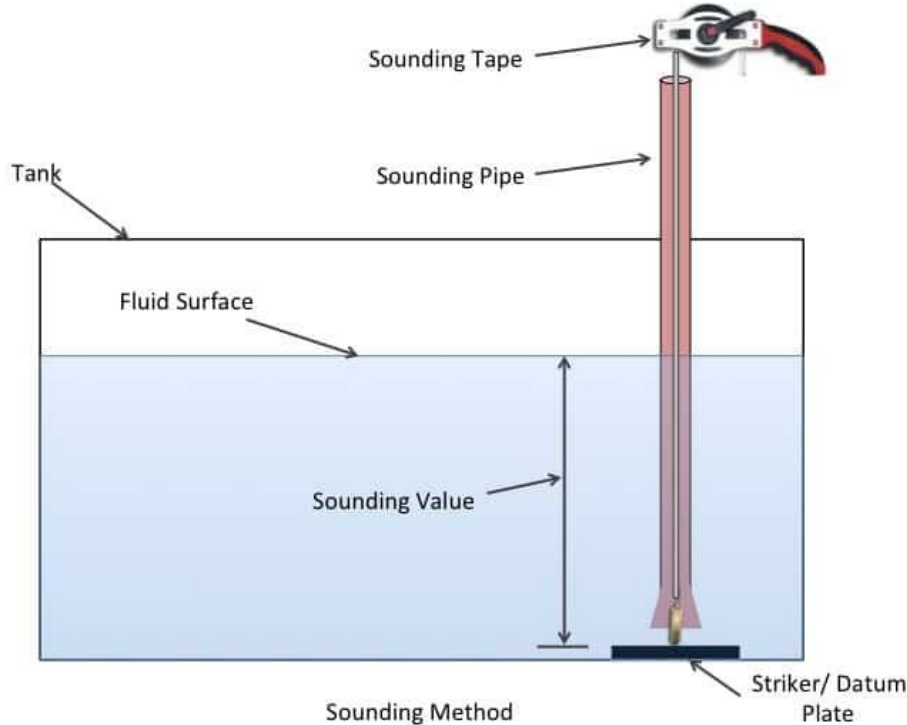
Trim by Head

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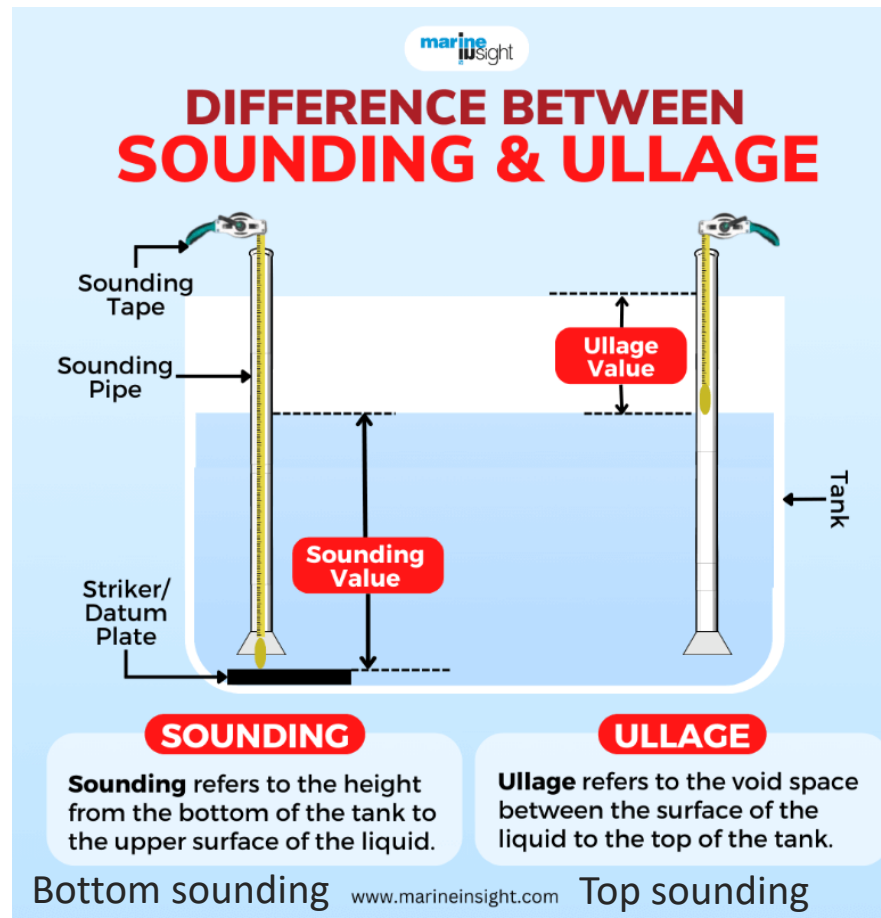
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



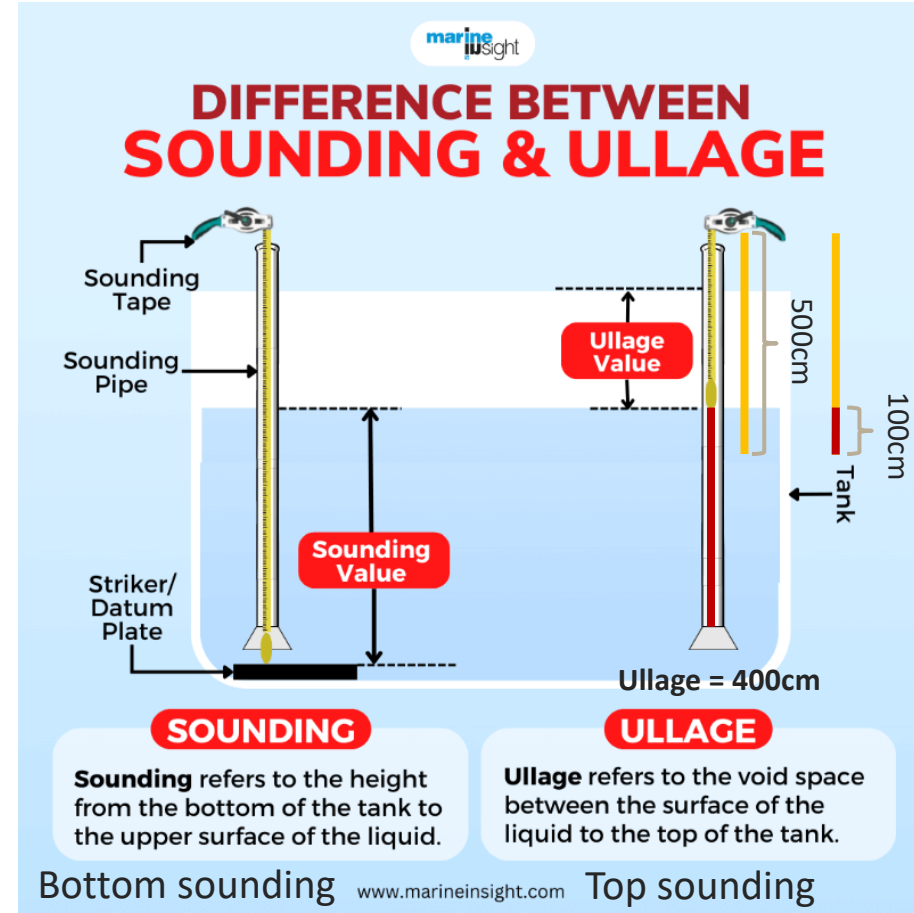
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 1)

Sounding depth = 856 cm / Trim = By the stern -3.0 m

How much is the volume of tank => 537.34 m³

Quiz 2)

Ullage depth = 463 cm / Trim = Even keel

How much is the volume of tank => 521.02 m³

Always sounding depth + ullage depth is same.

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN				ULLAGE DEPTH CM
			-1.0M	-2.0M	-3.0M	-4.0M	
800.	519.12	514.41	509.70	504.99	500.28	495.57	473.
802.	520.44	515.73	511.02	506.31	501.60	496.89	471.
804.	521.76	517.05	512.34	507.63	502.92	498.21	469.
806.	523.09	518.38	513.67	508.96	504.25	499.54	467.
808.	524.41	519.70	514.99	510.28	505.57	500.86	465.
810.	525.73	521.02	516.31	511.60	506.89	502.19	463.
812.	527.06	522.35	517.64	512.93	508.22	503.51	461.
814.	528.38	523.67	518.96	514.25	509.54	504.83	459.
816.	529.71	525.00	520.29	515.58	510.87	506.16	457.
818.	531.03	526.32	521.61	516.90	512.19	507.48	455.
820.	532.35	527.64	522.93	518.22	513.51	508.80	453.
822.	533.68	528.97	524.26	519.55	514.84	510.13	451.
824.	535.00	530.29	525.58	520.87	516.16	511.45	449.
826.	536.33	531.62	526.91	522.20	517.49	512.78	447.
828.	537.65	532.94	528.23	523.52	518.81	514.10	445.
830.	538.97	534.26	529.55	524.84	520.13	515.42	443.
832.	540.30	535.59	530.88	526.17	521.46	516.75	441.
834.	541.62	536.91	532.20	527.49	522.78	518.07	439.
836.	542.94	538.23	533.52	528.81	524.10	519.40	437.
838.	544.27	539.56	534.85	530.14	525.43	520.72	435.
840.	545.59	540.88	536.17	531.46	526.75	522.04	433.
842.	546.92	542.21	537.50	532.79	528.08	523.37	431.
844.	548.24	543.53	538.82	534.11	529.40	524.69	429.
846.	549.56	544.85	540.14	535.43	530.72	526.01	427.
848.	550.89	546.18	541.47	536.76	532.05	527.34	425.
850.	552.17	547.50	542.79	538.08	533.37	528.66	423.
852.	553.29	548.83	544.12	539.41	534.70	529.99	421.
854.	554.23	550.15	545.44	540.73	536.02	531.31	419.
856.	554.99	551.47	546.76	542.05	537.34	532.63	417.
858.	555.59	552.80	548.09	543.38	538.67	533.96	415.
860.	556.00	554.12	549.41	544.70	539.99	535.28	413.
862.	556.24	555.44	550.73	546.02	541.31	536.60	411.
864.	556.31	556.31	552.03	547.33	542.62	537.90	409.
866.	556.31	556.31	553.17	548.53	543.84	539.14	407.
868.	556.31	556.31	554.13	549.66	545.03	540.35	405.
870.	556.31	556.31	554.92	550.70	546.14	541.52	403.
872.	556.31	556.31	555.53	551.64	547.20	542.63	401.
874.	556.31	556.31	555.96	552.50	548.21	543.70	399.
876.	556.31	556.31	556.22	553.28	549.15	544.74	397.
878.	556.31	556.31	556.30	553.96	550.05	545.72	395.
880.	556.31	556.31	556.31	554.56	550.87	546.66	393.
882.	556.31	556.31	556.31	555.08	551.64	547.57	391.
884.	556.31	556.31	556.31	555.49	552.34	548.41	389.
886.	556.31	556.31	556.31	555.82	553.00	549.22	387.
888.	556.31	556.31	556.31	556.08	553.59	550.00	385.
890.	556.31	556.31	556.31	556.23	554.12	550.71	383.
892.	556.31	556.31	556.31	556.30	554.60	551.39	381.
894.	556.31	556.31	556.31	556.31	555.02	552.03	379.
896.	556.31	556.31	556.31	556.31	555.39	552.62	377.
898.	556.31	556.31	556.31	556.31	555.68	553.16	375.

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 3)

Sounding depth = 861 cm / Trim = By the stern -1.0 m

How much is the volume of tank => $(549.41+550.73)/2 = 550.07 \text{ m}^3$

Quiz 4)

Sounding depth = 861 cm / Trim = By the stern -1.5 m

How much is the volume of tank =>

$$\frac{(549.41+544.70)/2 + (550.73+546.02)/2}{2} = 547.715\text{m}^3 \cong 547.72\text{m}^3$$

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN				ULLAGE DEPTH CM
			-1.0M	-2.0M	-3.0M	-4.0M	
800.	519.12	514.41	509.70	504.99	500.28	495.57	473.
802.	520.44	515.73	511.02	506.31	501.60	496.89	471.
804.	521.76	517.05	512.34	507.63	502.92	498.21	469.
806.	523.09	518.38	513.67	508.96	504.25	499.54	467.
808.	524.41	519.70	514.99	510.28	505.57	500.86	465.
810.	525.73	521.02	516.31	511.60	506.89	502.19	463.
812.	527.06	522.35	517.64	512.93	508.22	503.51	461.
814.	528.38	523.67	518.96	514.25	509.54	504.83	459.
816.	529.71	525.00	520.29	515.58	510.87	506.16	457.
818.	531.03	526.32	521.61	516.90	512.19	507.48	455.
820.	532.35	527.64	522.93	518.22	513.51	508.80	453.
822.	533.68	528.97	524.26	519.55	514.84	510.13	451.
824.	535.00	530.29	525.58	520.87	516.16	511.45	449.
826.	536.33	531.62	526.91	522.20	517.49	512.78	447.
828.	537.65	532.94	528.23	523.52	518.81	514.10	445.
830.	538.97	534.26	529.55	524.84	520.13	515.42	443.
832.	540.30	535.59	530.88	526.17	521.46	516.75	441.
834.	541.62	536.91	532.20	527.49	522.78	518.07	439.
836.	542.94	538.23	533.52	528.81	524.10	519.40	437.
838.	544.27	539.56	534.85	530.14	525.43	520.72	435.
840.	545.59	540.88	536.17	531.46	526.75	522.04	433.
842.	546.92	542.21	537.50	532.79	528.08	523.37	431.
844.	548.24	543.53	538.82	534.11	529.40	524.69	429.
846.	549.56	544.85	540.14	535.43	530.72	526.01	427.
848.	550.89	546.18	541.47	536.76	532.05	527.34	425.
850.	552.17	547.50	542.79	538.08	533.37	528.66	423.
852.	553.29	548.83	544.12	539.41	534.70	529.99	421.
854.	554.23	550.15	545.44	540.73	536.02	531.31	419.
856.	554.99	551.47	546.76	542.05	537.34	532.63	417.
858.	555.59	552.80	548.09	543.38	538.67	533.96	415.
860.	556.00	554.12	549.41	544.70	539.99	535.28	413.
862.	556.24	555.44	550.73	546.02	541.31	536.60	411.
864.	556.31	556.31	552.03	547.33	542.62	537.90	409.
866.	556.31	556.31	553.17	548.53	543.84	539.14	407.
868.	556.31	556.31	554.13	549.66	545.03	540.35	405.
870.	556.31	556.31	554.92	550.70	546.14	541.52	403.
872.	556.31	556.31	555.53	551.64	547.20	542.63	401.
874.	556.31	556.31	555.96	552.50	548.21	543.70	399.
876.	556.31	556.31	556.22	553.28	549.15	544.74	397.
878.	556.31	556.31	556.30	553.96	550.05	545.72	395.
880.	556.31	556.31	556.31	554.56	550.87	546.66	393.
882.	556.31	556.31	556.31	555.08	551.64	547.57	391.
884.	556.31	556.31	556.31	555.49	552.34	548.41	389.
886.	556.31	556.31	556.31	555.82	553.00	549.22	387.
888.	556.31	556.31	556.31	556.08	553.59	550.00	385.
890.	556.31	556.31	556.31	556.23	554.12	550.71	383.
892.	556.31	556.31	556.31	556.30	554.60	551.39	381.
894.	556.31	556.31	556.31	556.31	555.02	552.03	379.
896.	556.31	556.31	556.31	556.31	555.39	552.62	377.
898.	556.31	556.31	556.31	556.31	555.68	553.16	375.

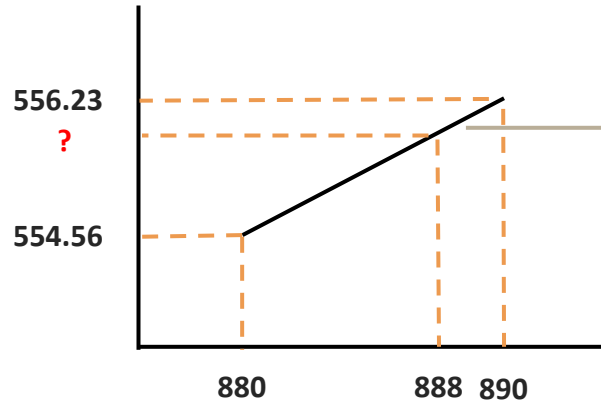
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 5)

Sounding depth = 888 cm / Trim = By the stern -2.0 m

How much is the volume of tank?



$$y = 0.167x + b$$

$$556.23 = 0.167 \times 890 + b$$

$$b = 407.6$$

$$y = 0.167 \times 888 + 407.6$$

$$y = 555.896$$

$$\text{Slope} = \frac{556.23 - 554.56}{890 - 880} = 0.167$$

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	TRIM BY THE STERN	TRIM BY THE STERN	TRIM BY THE STERN	TRIM BY THE STERN	TRIM BY THE STERN	ULLAGE DEPTH CM
			-1.0M	-2.0M	-3.0M	-4.0M	-5.0M	
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
860.	558.79	554.12	549.41	544.70	539.99	535.28	530.57	413.
870.	565.41	560.74	556.03	551.32	546.61	541.90	537.19	403.
880.	572.03	567.36	562.65	557.94	553.23	548.52	543.81	393.
890.	578.65	573.98	569.27	564.56	559.85	555.14	550.43	383.

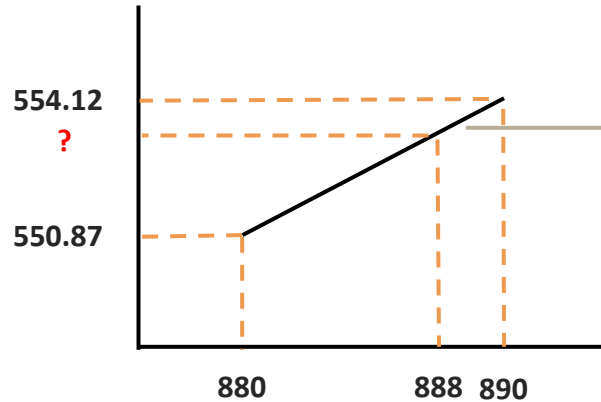
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 5)

Sounding depth = 888 cm / Trim = By the stern -3.0 m

How much is the volume of tank?



$$y = 0.325x + b$$

$$554.12 = 0.325 \times 890 + b$$

$$b = 264.87$$

$$y = 0.325 \times 888 + 264.87$$

$$y = 553.47$$

$$\text{Slope} = \frac{554.12 - 550.87}{890 - 880} = 0.325$$

*** TANK SOUNDING TABLES ***								
SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS					ULLAGE DEPTH CM
			TRIM BY THE STERN					
			-1.0M	-2.0M	-3.0M	-4.0M	-5.0M	
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.

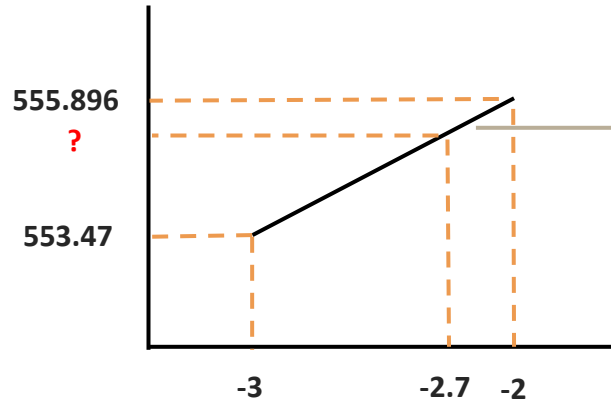
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 5)

Sounding depth = 888 cm / Trim = By the stern -2.7 m

How much is the volume of tank?



$$\text{Slope} = \frac{555.896 - 553.47}{-2 - (-3)} = 2.426$$

$$y = 2.426x + b$$

$$553.47 = 2.426 \times (-3) + b$$

$$b = 560.748$$

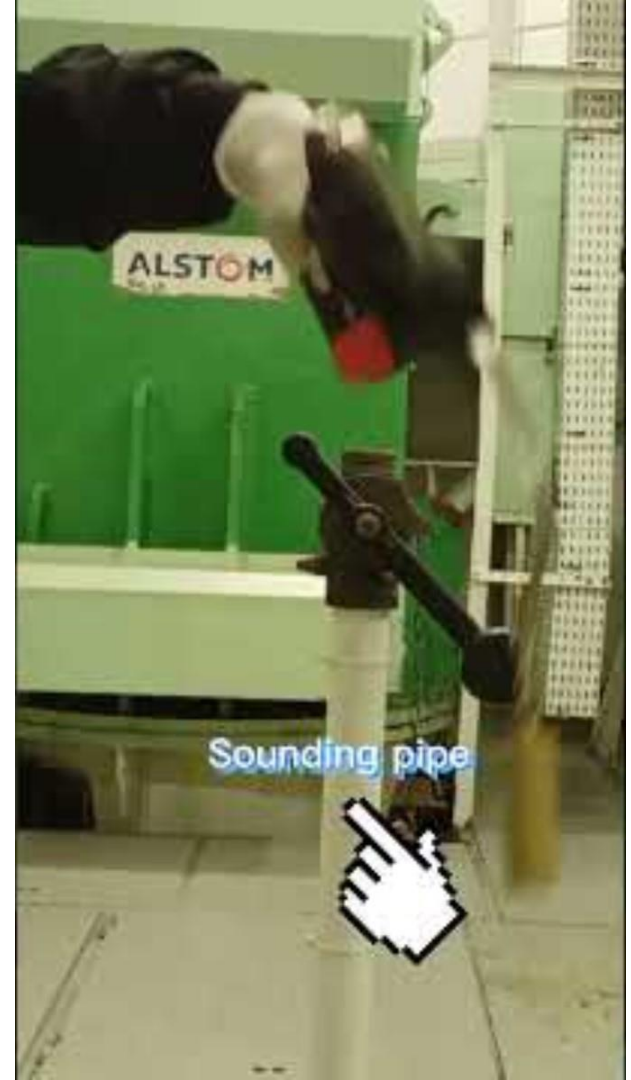
$$y = 2.426 \times (-2.7) + 560.748$$

$$y = 554.1978$$

*** TANK SOUNDING TABLES ***								
SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS					ULLAGE DEPTH CM
			TRIM BY THE STERN					
			-1.0M	-2.0M	-3.0M	-4.0M	-5.0M	
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



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Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



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Apprentice engineer (A/E) Duties

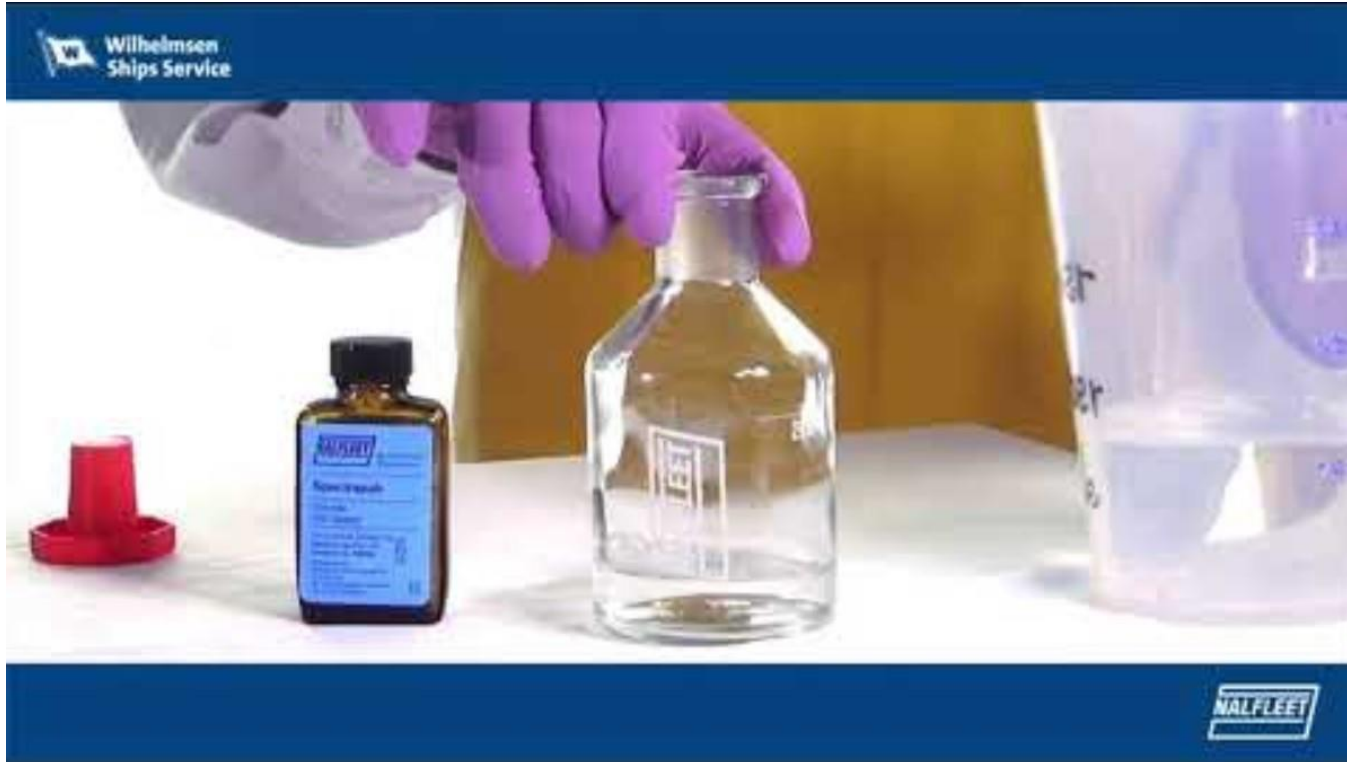
Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



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Apprentice engineer (A/E) Duties

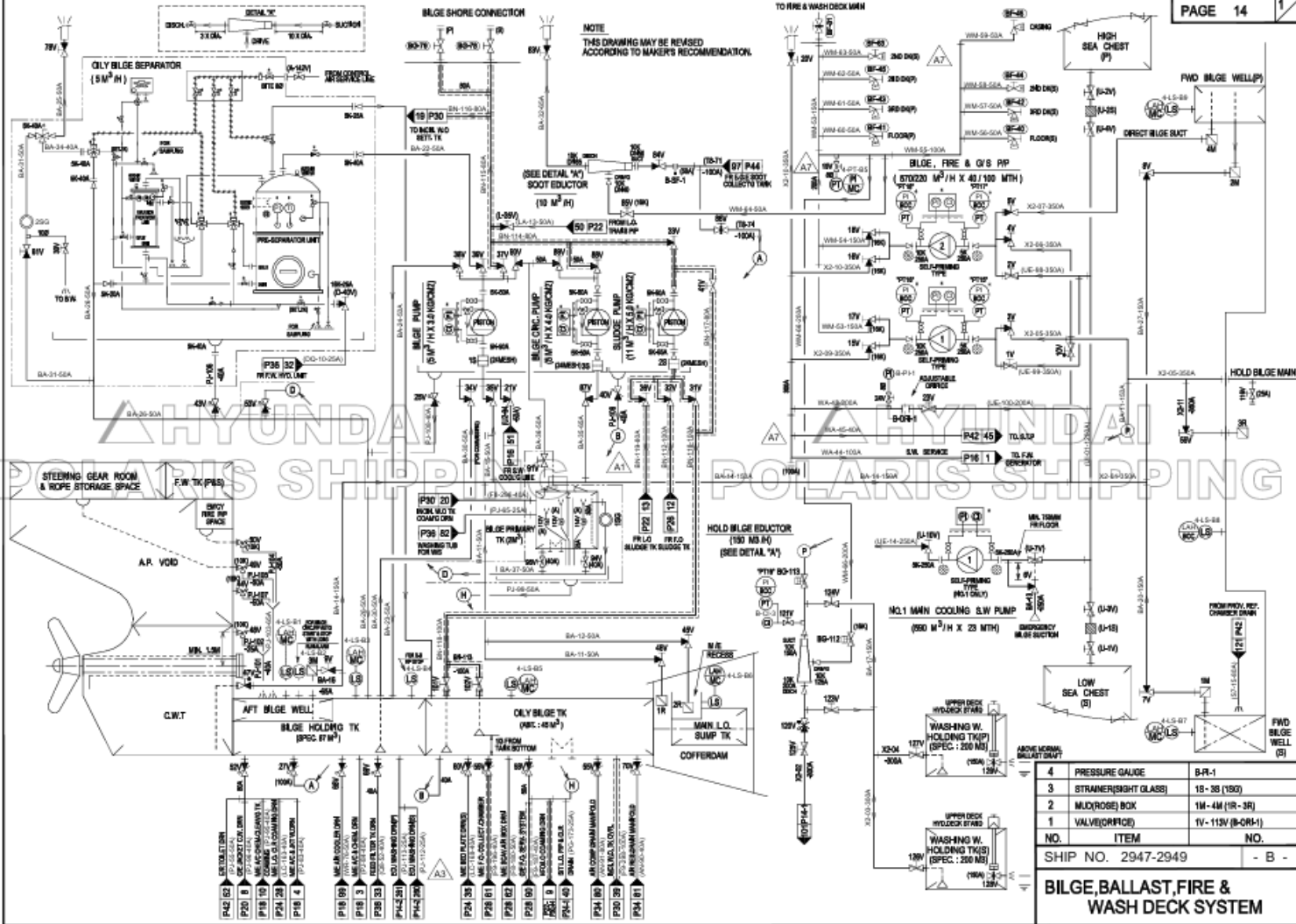
Cooling fresh water test (Please refer to another ppt file) & Boiler water test



Apprentice engineer (A/E) Duties

Transfer of bilge water from the bilge well to the bilge holding tank





NO.	ITEM	NO.
4	PRESSURE GAUGE	B-R-1
3	STRAINER(SIGHT GLASS)	18-35 (180)
2	MICROBE BOX	1M-4M (1R-3R)
1	VALVE(OFFICE)	1V-115V (B-OR-1)
SHIP NO. 2947-2949		- B -
BILGE, BALLAST, FIRE & WASH DECK SYSTEM		



Apprentice engineer (A/E) Duties

Transfer of bilge water from the bilge well to the bilge holding tank



Apprentice engineer (A/E) Duties



Apprentice engineer (A/E) preparation materials

You can prepare the materials by referring to the Excel file uploaded to GitHub and add any additional items you need.

승선 시 준비물

	NAME	QTY	REMARK	BUY	CHECK
1	근무복	1	하복/등복 상하의, 벨트		
2	체육복	1	볼가을용, 여름용		
3	필기구	N	필통, 펜, 사프, 화이트, 형광펜		
4	LED 손전등	1	손전등은 한 손에 잡히고 주머니에 넣을 수 있는 크기, 배터리, 충전기, 충전 케이블		
5	전자손목시계	1	Flowmeter를 통한 기름 계산할 때 필요함. 초시계 셀 수 있는 것. "카시오 F-91W-1D" 추천		
6	외장하드	1	4TB 정도면 충분		
7	USB	1			
8	개인서적	N			
9	실습 리포트 준비물	N			
10	반팔	6	일상용 : 4개, 상복용 : 2개		
11	바지	4	반바지 1개, 일반바지 2개		
12	수건	5			
13	세면도구	1	치약 1개, 칫솔 2개, 치간칫솔 n개		
14	선글라스	1			
15	양말	10	검은색 : 2개, 일반 : 8개		
16	운동화	1	승선할 때 신고 가면 됨		
17	손톱깎이	1			
18	전기면도기	1	충전기 같이 챙기기		
19	의약품	N	본인에게 꼭 필요한 것만 챙기기		
20	노트북	1	전원 선, 마우스, 노트북 가방		
21	핸드폰	1	충전기 같이 챙기기		
22	이어폰	1			
23	립크루즈	1			
24	손 수첩	1	기관실에서 실습 시 메모용 (주머니에 들어갈 사이즈) "양지사 피디수첩" 추천		
25	필기 노트	1			
26	지갑	1			
27	속옷	8	일반용 : 5, 작업복 입을 시 : 3		
28	안대	1			
29	루서	1			

BUY: 아직 준비가 되지 않아 구매가 필요한 것
CHECK: 준비가 완료되어 캐리어 안에 집어 넣은 것



Future career direction

First of all, what I'm writing here isn't the definitive answer to your career path. There are certainly better options out there, of which I'm unaware. So, please use the following information as a reference only.

Students majoring in marine systems engineering at KMOU are divided into four major groups after graduation.

- 1) Onboard the ship (Majorly for military service)
- 2) Military enlistment (ROTC)
- 3) Get a job at a company (Women and men who served in the military)
- 4) Graduate school admission (Women and men who served in the military)



Future career direction

1) Onboard the ship (Majorly for military service)

If you are boarding a ship, it is recommended to try to become a 1st engineer if possible.

The condition for obtaining a 1st class engineer license is to board a ship for one year as a 1st engineer while holding a 2nd class engineer license.

Therefore, you can meet the requirements by boarding a ship for about 5 to 6 years.

However, if given the opportunity, you can challenge yourself to become a chief engineer.

These days, promotions are quick, so there are cases where you can be promoted to chief engineer in your early 30s.

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Shipping companies (Where you worked or elsewhere)
- Shipbuilding companies
- Shipbuilding and marine equipment companies & Others
- Classification societies & Flag registry
- Ministry of Oceans and Fisheries
- Coast Guard
- Public institutions and corporations

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Shipping companies (Where you worked or elsewhere)



Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Shipbuilding companies

➤ HD한국조선해양

➤ HD현대중공업

➤ HD현대삼호

➤ HD현대마린솔루션

➤ HD현대일렉트릭

➤ HD현대사이트솔루션

➤ HD현대인프라코어



Future career direction

1) Onboard the ship (Majorly for military service)

Shipbuilding and marine equipment companies & Others

Everllence

WIN GD
Winterthur Gas & Diesel

PANASIA

DongHwa Entec
(주)동화엔텍


WÄRTSILÄ

ABB

mtu



Rolls-Royce®

DOOSAN 두산에너지빌리티


JONGHAP
(주) 종합해사
MARITIME INC.

LAB021


mapsea


marineworks



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Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Classification societies & Flag registry



Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Ministry of Oceans and Fisheries  해양수산부



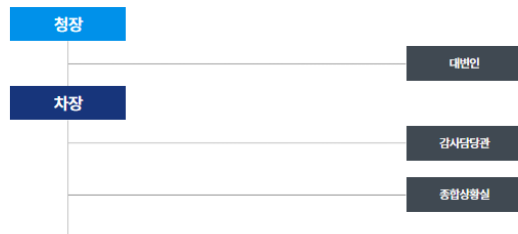
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Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer,
you can challenge yourself for the following jobs.

- Coast Guard



운영지원과	기획조정관	경비국	구조안전국	수사국	정보외사국	해양오염방제국
	기획재정담당관 혁신행정법무담당관 인사담당관 교육훈련담당관 국제협력담당관 인공지능전담팀	경비직전과 항공과 해상교통관제과 마개경비기획과	해양안전과 수색구조과 수상레지과	수사기획과 수사심사과 형사과 과학수사팀 중대범죄수사팀	정보과 외사과 보안과 장비기술국 장비기획과 장비관리과 정보통신과	병제기획과 기동방제과 해양오염예방과

해양경찰교육원	중앙해양특수구조단	해양경찰장비상	제주지방해양경찰청	서울지방해양경찰청	남해지방해양경찰청	동해지방해양경찰청
해양경찰연구센터	서해해양특수구조대 동해해양특수구조대	해양경찰부산장비상	인천해양경찰서 평택해양경찰서 태안해양경찰서 보령해양경찰서 서해5도특별경비단	군산해양경찰서 부안해양경찰서 목포해양경찰서 연도해양경찰서 여수해양경찰서	동명해양경찰서 장원해양경찰서 부산해양경찰서 울산해양경찰서 사천해양경찰서	포항해양경찰서 울진해양경찰서 동해해양경찰서 속초해양경찰서 강릉해양경찰서
			제주지방해양경찰청 제주해양경찰서 서귀포해양경찰서			



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Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Public institutions and corporations



Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Public institutions and corporations



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Korea Institute of Maritime and Fisheries Technology



한국조선해양기자재연구원
Korea Marine Equipment Research Institute



KOEM 해양환경공단



한국지질자원연구원
Korea Institute of Geoscience and Mineral Resources

KOMSA
한국해양교통안전공단
KOREA MARITIME TRANSPORTATION SAFETY AUTHORITY



기상청



국 방 과 학 연 구 소
Agency for Defense Development



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Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Public institutions and corporations



Future career direction

2) Military enlistment (ROTC)

After ROTC, you can develop your expertise by boarding a ship in shipping company, and consider other paths such as employment at a shipbuilding companies or the Coast Guard, or going on to graduate school.

If you board a ship after ROTC, the company will generally promote you quickly.

Long-term service in the military is also an option.

Because society sometimes gives preference to ROTC graduates, you can apply to various companies in addition to the maritime companies mentioned above.

Future career direction

3) Get a job at a company (Women and men who served in the military)

If you apply for a job right after graduating from university, you can apply to a variety of fields just like students at regular universities.

However, applying to a company related to marine engineering is very advantageous compared to other general university students.

Companies that put up job-related placards and posters at university can be considered to be favorable to the university.

When applying to companies in different fields than ours, you will need to prepare specifications that are appropriate for the position.

Future career direction

4) Graduate school admission

If finding a job is difficult, graduate school may be an alternative option.

However, I personally recommend that you look for a job after graduating with a master's or doctorate degree rather than looking for a job while attending graduate school.

Going to graduate school after boarding the ship is also a good career path.

The best reason to apply for graduate school is to have a clear sense of purpose about your career path after graduate school. Because, during graduate school, you have to sit in a chair and focus on research most of the time, so it is a different environment from usual, and you also have to enjoy research, have many new ideas, and have a challenging attitude.

Future career direction

4) Graduate school admission

Currently, the competition rate for admission to KMOU's graduate school is somewhat high.

This is because a high percentage of working students (=currently working at company) choose to pursue graduate school part-time. However, if you pursue graduate school full-time, your chances of admission significantly increase.

When choosing a graduate school, it's important to confirm whether the professor's research area aligns with your desired direction. This can be determined by reviewing the papers listed on the professor's website. (Carefully check the number and quality of papers written by the professor as the first author. Identify the first author rather than co-authors and corresponding authors.)

Even if it's not KMOU, there are many excellent graduate schools in Korea and abroad, so it's not a bad idea to think about it from various perspectives.



Future career direction

4) Graduate school admission

Below are some research positions that I believe you might be interested in pursuing with a marine systems engineering degree. Please note that not all positions are open to you; only positions that align with your major are eligible.



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Agency for Defense Development



KITECH
한국생산기술연구원



한국해사협력센터
Korea Maritime Cooperation Center



한국조선해양기자재연구원
Korea Marine Equipment Research Institute



한국해양과학기술원



한국해양수산개발원
KOREA MARITIME INSTITUTE



한국기계연구원
KOREA INSTITUTE OF MACHINERY & MATERIALS



중소조선연구원



한국전자통신연구원
Electronics and Telecommunications
Research Institute



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한국에너지기술연구원
KOREA INSTITUTE OF ENERGY RESEARCH



한국전기연구원
KOREA ELECTROTECHNOLOGY RESEARCH INSTITUTE



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Future career direction

A challenge that takes time but is worth it

- 1) Coast Guard officer candidate
- 2) Maritime law schools (Korea University, Jeju National University, etc.)
- 3) National Intelligence Service, Grade 7
- 4) Ministry of Oceans and Fisheries, Grade 5

References

- 1) <https://www.youtube.com/watch?v=aWeAaHFV25w>
- 2) <https://www.primetools.co.uk/product/facom-44-ju6-af-open-end-wrench-set/>
- 3) <https://garageshop24x7.com/12pc-double-open-end-spanners-set-6x7-to-30x32mm/?srsltid=AfmBOorc6wUV8ANn6syulRVrDQbgVkRkvlfZsEHtVrhq0ItHdQQ6ag5o>
- 4) https://www.eastmanshop.com/products/eastman-bi-hexagonal-ring-spanner-sets-chrome-vanadium-steel-chrome-plated-box-pack-e-2007?srsltid=AfmBOopvWl6whMOhsHTvSyUSkNscTxMuPlwFCT_oT35DnZoG6N-TcfMY
- 5) <https://www.indiamart.com/proddetail/ring-spanner-22208178162.html>
- 6) <https://kr.rs-online.com/web/p/spanners/1883835>
- 7) <https://www.drapertools.com/product/97918/draper-hi-torq-metric-combination-spanner-set-6-32mm-25-piece/>
- 8) <https://www.youtube.com/shorts/nM9rTiprrSc>



References

- 9) <https://www.youtube.com/watch?v=KHKFy0J75Mo>
- 10) <https://www.youtube.com/watch?v=0aJbnn8CWgg>
- 11) <https://kr.rs-online.com/web/p/spanners/7179033>
- 12) <https://www.piccu.org/product-p-187760.html>
- 13) https://www.asahi-tool.co.jp/english/product/hook-spanner/cate-fk-hook-spanner_english.html
- 14) <https://int.rsdelivers.com/product/skf/hn-4-16/set/skf-hn-series-9-piece-spanner-set-hn-4-hn-16-mm/1248475>
- 15) <https://www.maxclawtools.com/adjustable-hook-spanner-wrench-hw-311.html>
- 16) <https://www.maxiloc.com.au/products/industrial-supplies/hand-tools/hook-wrenches-c-spanners-pin-spanners/>



References

- 17) <https://www.youtube.com/watch?v=HO3qmh3Yd1c>
- 18) <https://daitool.com/blogs/daitool-blog/the-best-guide-to-removing-a-stripped-bolt?srsId=AfmBOoolZLAGFicdBjDP42h3RDLArJTA0R7dBG1GBNztNL3plfjOsISE>
- 19) <https://www.tiktok.com/@stephencurnow/video/7000139401238023429>
- 20) <https://worstroom.com/how-to-remove-a-stripped-bolt/>
- 21) <https://cartreatments.com/how-to-remove-stripped-bolt/>
- 22) https://www.reddit.com/r/3rdGen4Runner/comments/136vata/stripped_one_of_my_lower_ball_joint_bolts_trying/
- 23) <https://www.britannica.com/technology/bolt-fastener>
- 24) <https://www.iqsdirectory.com/articles/bolts/types-of-bolts.html>
- 25) <https://www.indiamart.com/proddetail/mild-steel-nut-bolt-9642377530.html?srsId=AfmBOoo4exEEP-RhKDJsiKeo6vkHiliJsGJWVvejK-Y-mswY8qGx4qsY>

References

- 26) <https://kr.rs-online.com/web/p/pliers/9103007>
- 27) <https://www.knipex.com/products/pipe-wrenches-and-water-pump-pliers/pipe-wrenches-90/pipe-wrenches/8310010>
- 28) <https://www.youtube.com/shorts/HvDnD6LcHaY>
- 29) <https://www.youtube.com/shorts/J-VxJSGRKD4>
- 30) https://ko.wikipedia.org/wiki/%EC%86%8C%EC%BC%93_%EB%A0%8C%EC%B9%98
- 31) <https://www.kincrome.com.au/sockets-socket-sets-accessories/socket-%28standard%29-7mm-1-4-drive-kc457c>
- 32) <https://www.amazon.com/Olsa-Tools-6-Point-Mechanic-Professional/dp/B0921J17C2?th=1>
- 33) <https://www.indiamart.com/proddetail/drive-socket-wrench-set-25899754433.html>
- 34) <https://www.youtube.com/shorts/7K3ubtTfcX4>

References

- 35) <https://www.youtube.com/shorts/LqzGjqxXX54>
- 36) https://www.youtube.com/shorts/HZGTVJfq_SY
- 37) <https://www.youtube.com/watch?v=7LsKxc97Nxs>
- 38) <https://www.youtube.com/shorts/2t05UblrWjE>
- 39) <https://www.ubuy.com.bh/en/product/2F6S7PJA-kseibi-211251-t-handle-socket-wrench-set-6-point-socket-7-pack?srsId=AfmBOoprYkCWfHpa5nvbaWG1vePkOa0lpWbDivx9GUOqN9KjJqZylSph>
- 40) <https://www.amazon.in/SAENTE-13-Piece-T-Handle-Socket-Set/dp/B0F59L4Y6X>
- 41) <https://www.dillonsupply.com/Product/ridgid-31330?srsId=AfmBOopViN8-3otOk797kg7wlfowNxn2JnEDdXkOXUzS6ecW-fSN0tTZ>
- 42) <https://www.gigatools.ph/products/ridgid-chain-wrenches>
- 43) <https://www.popularmechanics.com/home/tools/a4114/how-to-use-an-allen-wrench/>

References

- 44) https://en.wikipedia.org/wiki/Hex_key
- 45) <https://www.amazon.ca/InPower-Wrench-Metric-Socket-Screws/dp/B0921N1BZV>
- 46) <https://www.rhhardware.com/ko/AW-Carbon-Steel-Hex-Key-Allen-Wrench-1-5mm-to-22mm-pd852835.html>
- 47) <https://en.wikipedia.org/wiki/Hexagon>
- 48) <https://www.fas10.in/allen-bolts/>
- 49) <https://www.facebook.com/interior76/posts/allen-bolt-size-vs-allen-key-size-/953264849595963/>
- 50) <https://yibangfastener88.en.made-in-china.com/product/gFqTDIkvEptd/China-DIN912-Hex-Socket-Head-Bolt-12-9-Lszx-Head-Mark-Allen-Bolt-Hexagon-Screws-Full-Thread-Hexagon-Head-Bolts.html>
- 51) <https://www.amazon.com/TEKTON-Drive-Socket-19-Piece-SHB91315/dp/B0BZX18KR7?th=1>
- 52) <https://www.amazon.com/Mayouko-Ball-End-Socket-3-10mm-Metric/dp/B0C8MN37RW>



References

- 53) <https://neikotools.com/products/neiko-01137b-1-2-drive-allen-bit-socket-set-8-piece-sae-1-4-to-3-4-3-length-cr-mo-impact-grade>
- 54) <https://www.harborfreight.com/9-piece-38-and-12-drive-metric-hex-bit-socket-set-67880.html>
- 55) <https://www.wonkeedonkeetools.co.uk/sockets-and-socket-sets/what-is-a-hex-socket>
- 56) https://en.wikipedia.org/wiki/Impact_wrench
- 57) https://www.agriculture.com/machinery/tools/power-tool/air-impact-wrenches-buyers-guide_248-ar45140
- 58) <https://autoedu.info/en/equipment-and-tools/tools/air-tools/air-impact-wrench/>
- 59) https://www.specialized.net/irwin-vise-grip-7wr-curved-jaw-locking-pliers-7.html?srsId=AfmBOoqyGTs5QeMHM42144k_JvVNmXUVU6oDH4QLwm8w8ppJDcRV0E4T
- 60) <https://www.dakotariggers.com/products/the-original%E2%84%A2-irwin-vise-grip-curved-jaw-pliers>
- 61) https://www.alibaba.com/product-detail/Ratchet-Spanner-Wrench-For-Installing-The_60727040632.html

References

- 62) <https://www.amazon.ca/HOIGON-Scaffold-Ratchet-Aligning-Scaffolding/dp/B095M2Y8H1>
- 63) <https://kr.rs-online.com/web/p/spanners/0487040>
- 64) <https://www.indiamart.com/proddetail/combo-ratchet-spanner-flexible-type-24745222012.html?srsltid=AfmBOorvxSlZ2pAbG3y4RqgS8Xpjb52PTdxtJdpDUxWBKKPIH95O9iDC>
- 65) <https://kr.rs-online.com/web/p/spanners/8764913>
- 66) <https://www.thetoolacademy.com/teng-6503rx-3-piece-multidrive-ratchet-ring-spanner-set-57707-p.asp>
- 67) <https://cnzengo.en.made-in-china.com/product/CZBfPdlyGFcn/China-Refrigeration-Hand-Tools-Ratchet-Wrench-CT-122-CT-123L-Ratchet-Spanner.html>
- 68) https://www.indiamart.com/proddetail/slugging-wrenches-striking-wrenches-2851625306997.html?srsltid=AfmBOoqMggfq2kn3uSi-jpVBe2wHEZQwqw3ojhUjB_UcswCea0n_GKrB
- 69) <https://www.amazon.com/uxcell-Striking-Wrench-Tensile-Spanner/dp/B0BNHCGF4R?th=1>
- 70) <https://www.indiamart.com/proddetail/wrench-striking-open-2851273729812.html?srsltid=AfmBOopMHbDVPpsW3fYqbeWqo9TXe828IO6uqfHZmmTnbPa0iFLliRMMy>

References

- 71) <https://www.amazon.com/uxcell-Striking-Wrench-Tensile-Spanner/dp/B0BNHCGF4R?th=1>
- 72) <https://www.indiamart.com/proddetail/wrench-striking-open-2851273729812.html?srsId=AfmBOopMHbDVPpsW3fYqbeWqo9TXe828IO6uqfHZmmTnbPa0iFLiRMMy>
- 73) <https://hultafors.com/en-gl/products/copper-hammer-cu>
- 74) <https://kr.rs-online.com/web/p/mallets/2215128>
- 75) <https://shopee.co.id/search?keyword=mallet&page=0>
- 76) https://www.bahco.com/int_en/stainless-steel-sledge-hammers-with-fibreglass-handle-pb_ss502_.html
- 77) https://www.stahlwerk-schweissgeraete.de/hammer-300g-en?srsId=AfmBOoqOb0kvE_jv8pvKFF3Tbt-5P4XjcPCyTRDGnovW2cW08YIpuFH-
- 78) <https://impa-catalogue.shipserv.com/59-pneumatic-electrical-tools/pneumatic-hand-tools/pneumatic-derusting-brushes>
- 79) <https://www.chutuomarine.com/pneumatic-derusting-brush-product/>



References

- 80) <https://www.garagesanctum.com/size-chart/bolt-spanner-size-chart-pdf/>
- 81) <https://www.renoximpex.com/blog/guidance-on-spanner-size-chart-for-bolts/>
- 82) <https://www.facebook.com/groups/engineeringeducationalstuff/posts/1861959907945064/>
- 83) <https://fittertraining.com/piping-and-structural-bolt-spanner-size-chart-stud-thread-mm-size-to-spanner-size-chart/>
- 84) <https://amesystem.com.au/products/hexagon-head-bolt-hhv-m8-bolts>
- 85) <https://kr.rs-online.com/web/p/pliers/8755809>
- 86) <https://kr.rs-online.com/web/p/pliers/8755802>
- 87) <https://kr.rs-online.com/web/p/pliers/7276702>
- 88) <https://www.faithfulltools.com/p/FAIPLDC6N/Diagonal-Cutting-Pliers-160mm>



References

- 89) <https://glasstools.co.nz/diagonal-cutting-pliers-side-cutters.html>
- 90) <https://kr.rs-online.com/web/p/pliers/2483722>
- 91) <https://www.sunshinegaskets.com/products/16-pcs-hollow-hole-gasket-punching-tool-set.html>
- 92) <https://www.mercadolibre.com.co/round-disc-cutting-kit-manual-round-punch-for-decals-craft/p/MCO2015760038>
- 93) <https://www.canapechic.com/product-p-539172.html>
- 94) <https://lotustools.ph/products/lotus-hollow-punch-set-9pc-lt900hpx>
- 95) <https://www.tiktok.com/@engineerroominsight/video/7535494395844463879>
- 96) <https://www.youtube.com/shorts/7yXlQ2EvFZM>
- 97) <https://www.youtube.com/watch?v=VTKPlcUAPt4>



References

- 98) <https://www.youtube.com/watch?v=0BtXgE6NSYE>
- 99) <https://kr.element14.com/duratool/dt000364/adj-torque-wrench-carbon-steel/dp/2918433>
- 100) <https://www.wikihow.com/Use-a-Torque-Wrench>
- 101) <https://www.mechdaily.com/what-is-hacksaw/>
- 102) <https://kr.rs-online.com/web/p/hand-saws/4499577>
- 103) <https://www.theengineeringchoice.com/what-is-file-tool/>
- 104) https://learninglab.rmit.edu.au/Toolbox/electrotech/toolbox1204/resources/03workshop/05hand_tools/08files.htm
- 105) <https://www.amazon.com/Ready-Abrasive-120grid-180grid-240grid/dp/B09F56FCMH?th=1>
- 106) <https://makerbazar.in/products/sandpaper-waterproof-abrasive-paper?srsltid=AfmBOooZPMcNvCyEd5JyGUsUsbhhjAViqQ9peYKMTgKU2eMcN68BkzB5>

References

- 107) <https://kr.rs-online.com/web/p/wire-strippers/0663617>
- 108) <https://major-tech.com/product/1-0mm-3-2mm-automatic-wire-stripper?srsId=AfmBOoo04W77jZ0UMZZRyMXhac2V4FhWZgE4G0R86VJBsKlitmxzCCzM>
- 109) <https://www.youtube.com/shorts/kUdPVFD7RuU>
- 110) <https://www.12voltplanet.co.uk/copper-tube-ratchet-crimping-tool-1-5-10mm2.html?srsId=AfmBOorm7Vur-ju83OG7w64adqCeLxPnUAQKTyIINVclODGM1t4SIEMM>
- 111) <https://m.vctec.co.kr/product/%EB%A7%88%EC%9D%B4%ED%81%AC%EB%A1%9C-%EC%BB%A4%EB%84%A5%ED%84%B0-%ED%81%AC%EB%A6%BC%ED%95%91-%ED%8C%8C%EC%9D%BC%EB%9F%AC-pa-24-micro-connector-crimping-pliers-pa-24/21601/>
- 112) <https://www.youtube.com/watch?v=BaKmAT9ipuU>
- 113) <https://www.huyett.com/blog/spring-washers?srsId=AfmBOopowDIAqTCE-ds8jVhXf5eVhWhCCeSUiHEX-sUSycZ0vj8OCicy>

References

- 114) <https://kr.element14.com/tr-fastenings/dm2-5-sra2was100din127/spring-washer-ss-a2-2-9mm-5-1mm/dp/2506051>
- 115) <https://www.asmc.net/1-1-2-flat-washer-sae-pattern-low-carbon-steel-zinc-plated/>
- 116) https://www.ubuy.kr/en/product/1FDFPETS-1-4-stainless-flat-washer-5-8-outside-diameter-304-stainless-steel-washers-flat-80-pcs?srsltid=AfmBOoqDlrv8LVD226LG0GyeRuLk5UH2R1TvUazS50_OvvmU-l4ya66Y&ref=hm-google-redirect
- 117) <https://kr.rs-online.com/web/p/circlips/0289247>
- 118) <https://itafasteners.com/products-clips-internal-circlip.php>
- 119) <https://plierpincer.com/what-is-circlips/>
- 120) <https://www.trfastenings.com/products/Catalogue/Washers-Circlips-Rings-Seals-O-Rings/Circlips/E-Type-External-Circlips>
- 121) <https://ko.aliexpress.com/item/32846318625.html?gatewayAdapt=glo2kor>



References

- 122) <https://kr.rs-online.com/web/p/pliers/5416797>
- 123) <https://www.amazon.in/GIZMO-Internal-Internnal-Straight-External/dp/B09DX9TPLX>
- 124) <https://www.youtube.com/shorts/SgeOBilWsdA>
- 125) <https://www.youtube.com/shorts/wdONQR4nc04>
- 126) <https://www.amazon.sa/-/en/Multimeter-Voltmeter-Multifunction-Backlight-Resistance/dp/B0BGLC5PJJ>
- 127) <https://www.bftb.org/product-p-313999.html>
- 128) <https://www.ebay.ca/itm/226279395191>
- 129) <https://www.ebay.com/itm/305974697583>
- 130) <https://www.tradeindia.com/products/generator-cylinder-head-c5222135.html>

References

- 131) <https://www.sciencedirect.com/science/article/pii/S135063071831553X>
- 132) <https://muckymariners.com/app/meo-class-4-subjects/function-4b-meol/engines/10-cylinder-head/cylinder-head-4-5-hydraulic-jack-unit/>
- 133) <https://www.youtube.com/watch?v=PFuJb9ollyY>
- 134) <https://www.youtube.com/watch?v=EsSRvF3e9ew>
- 135) <https://www.youtube.com/shorts/XdhcfXivST4>
- 136) <https://gdhygiene.com/product/cotton-work-gloves/>
- 137) <https://www.abtec4abrasives.com/150-mm-heavy-duty-bench-grinder--wire-wheel-450-watt-sealey-bg150xw99-3143-p.asp>
- 138) https://www.reddit.com/r/Machinists/comments/l4t7q9/guy_at_work_had_a_boo_boo_on_a_bench_grinder_said/



References

139) <https://www.youtube.com/watch?v=Lold81blCw8>

140) <https://www.skipper.co.uk/catalogue/item/brown-s-chief-engineer-s-log-book-no-132>

141) [https://mdnautical.com/brown-son-ferguson/21838-chief-engineer-s-log-book-no-132.html#googtrans\(en|ko\)](https://mdnautical.com/brown-son-ferguson/21838-chief-engineer-s-log-book-no-132.html#googtrans(en|ko))

142) <https://www.marinesocietyshop.org/engine-log-book>

143) https://www.amnautical.com/products/chief-engineers-log-book-no-132-3-months?srsId=AfmBOoqa90jKJ_RT1sBMSCR4UQJG8LwxJhFGqT0xvQ1NEeS-dzkk9G6m

144) http://www.tekhar.com/Programma/HYUNDAI/pdf_pict/Ships_pruduction/Upravlenie_ships/ACONIS2000E_INTRODUCTION.pdf

145) https://www.youtube.com/watch?v=ThzuvO3mZ_E

146) <https://www.youtube.com/watch?v=2SNozduHhhl>

References

147) https://www.youtube.com/watch?v=ESTS_kKD6TY&t=11s

148) <https://www.youtube.com/watch?v=56UOZrtqR8c>

149) <https://www.youtube.com/watch?v=5QISxfC60gY>

150) <https://www.youtube.com/watch?v=AWgryOyjUJ0>

151) <https://www.youtube.com/shorts/SPz0SUIHiMA>

152) <https://www.youtube.com/watch?v=ALqAymCtltY>

153) <https://www.youtube.com/watch?v=lsqIZQ49-JA>

154) <https://www.youtube.com/watch?v=pkGXzGp4AeA>

References

155) <https://www.youtube.com/shorts/ZzfPLX-0WFw>

156)
<https://www.facebook.com/photo.php?fbid=5919154268108309&id=173058702717923&set=a.324659590891166>

157) https://www.researchgate.net/figure/The-trim-and-the-draft-of-a-vessel_fig1_364078770

158) <https://www.marineinsight.com/guidelines/how-and-why-to-take-manual-sounding-on-a-ship/>

159) <https://www.weitongmarine.com/products/impa-650858-650859-30m-50m-white-steel-oil-tank-sounding-tape>

160) <https://www.youtube.com/shorts/P6SEraTtbEo>

161) <https://www.marineinsight.com/guidelines/how-and-why-to-take-manual-sounding-on-a-ship/>

162) <https://www.shutterstock.com/search/ship-engine-room>



References

- 163) <https://www.youtube.com/shorts/AdBtVLiBacM>
- 164) https://www.youtube.com/shorts/sCPSwltV_NA
- 165) <https://www.youtube.com/watch?v=EGTwA1PWhOk>
- 166) <https://www.youtube.com/watch?v=0aAT2zq5WQg>
- 167) <https://www.youtube.com/watch?v=9c8V745sYYg>
- 168) <https://www.youtube.com/watch?v=SF2pDGF1JPY>
- 169) <https://chiefengineerlog.com/2022/07/09/bunker-tanks-soundings-and-fuel-quantity-calculation/>
- 170) <https://www.stxoceanservice.com/ko>



References

- 171) <http://www.maritimepress.co.kr/news/articleView.html?idxno=311001>
- 172) <https://www.newswire.ca/news-releases/new-ceo-for-wilhelmsen-ship-management-884091754.html>
- 173) <https://www.newswire.ca/news-releases/new-ceo-for-wilhelmsen-ship-management-884091754.html>
- 174) <https://www.tradlinx.com/bl-tracking/kmtc>
- 175) <https://namu.wiki/w/HMM>
- 176)
<https://www.teamblind.com/kr/company/%EC%97%90%EC%9D%B4%EC%B9%98%EB%9D%BC%EC%9D%B8%ED%95%B4%EC%9A%B4/>
- 177) <https://namu.wiki/w/%ED%98%84%EB%8C%80%EA%B8%80%EB%A1%9C%EB%B9%84%EC%8A%A4>
- 178) https://www.gmarineservice.com/html/00_main/

References

179) <http://www.possm.com/>

180) <https://www.polaris-shipping.com/>

181) <https://namu.wiki/w/SM%EC%83%81%EC%84%A0>

182) <https://www.schednet.com/adv/sinokor/index202010.asp>

183)
https://www.hanwha.co.kr/newsroom/media_center/digital_library/view.do?seq=8300&board=PRESS

184) <https://recruit.hd.com/kr/mainLayout/applyDetail/226764>

185) <https://namu.wiki/w/%ED%95%9C%ED%99%94%EC%97%94%EC%A7%84>

186) <https://www.youtube.com/@samsungshi>



References

187) <https://namu.wiki/w/%EC%BC%80%EC%9D%B4%EC%A1%B0%EC%84%A0>

188) <https://jasoseol.com/companies/13056>

189)
https://ko.wikipedia.org/wiki/%ED%8C%8C%EC%9D%BC:HJ%EC%A4%91%EA%B3%B5%EC%97%85_%EB%A1%9C%EA%B3%A0%28%ED%95%9C%EA%B8%80%29.png

190) <https://namu.wiki/w/HSG%EC%84%B1%EB%8F%99%EC%A1%B0%EC%84%A0>

191)
https://ko.wikipedia.org/wiki/%ED%8C%8C%EC%9D%BC:%ED%95%9C%EA%B5%AD%EC%84%A0%EA%B8%89_%EB%A1%9C%EA%B3%A0.JPG

192) <https://www.maritimelondon.com/member/abs-europe-ltd>

193) <https://www.roxtec.com/kr/jepum/injeungseo/classnk/>

194) <https://www.dnv.co.kr/>



References

195) <https://www.nidv.eu/en/industry-guide/lloyds-register-emea/>

196) https://en.wikipedia.org/wiki/RINA_%28company%29

197) https://ko.wikipedia.org/wiki/%EB%B7%B0%EB%A1%9C_%EB%B2%A0%EB%A6%AC%ED%83%80%EC%8A%A4

198)
<https://namu.wiki/w/%EA%B5%AD%EB%A6%BD%ED%95%B4%EC%96%91%EC%A1%B0%EC%82%AC%EC%9B%90>

199)
<https://namu.wiki/w/%EA%B5%AD%EB%A6%BD%EC%88%98%EC%82%B0%EA%B3%BC%ED%95%99%EC%9B%90>

200) <https://namu.wiki/w/%ED%95%B4%EC%96%91%EC%88%98%EC%82%B0%EB%B6%80>

201) <https://www.facebook.com/SouthSFMS/>

202) <https://eastship.mof.go.kr/ko/index.do>

References

203) <https://westship.mof.go.kr/ko/index.do>

204) <https://www.ust21.co.kr/>

205)
<https://namu.wiki/w/%EC%9A%B8%EC%82%B0%EC%A7%80%EB%B0%A9%ED%95%B4%EC%96%91%EC%88%98%EC%82%B0%EC%B2%AD>

206) <https://pohang.mof.go.kr/ko/index.do>

207) <https://yeosu.mof.go.kr/ko/index.do>

208)
<https://namu.wiki/w/%EB%8C%80%ED%95%9C%EB%AF%BC%EA%B5%AD%20%ED%95%B4%EC%96%91%EA%B2%BD%EC%B0%B0%EC%B2%AD>

209) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EA%B0%80%EC%8A%A4%EA%B3%B5%EC%82%AC>

210) <https://www.wemakenews.co.kr/news/view.php?no=19297>

References

211) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%A0%84%EB%A0%A5%EA%B8%B0%EC%88%A0>

212) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%A0%84%EB%A0%A5%EA%B3%B5%EC%82%AC>

213) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%88%98%EB%A0%A5%EC%9B%90%EC%9E%90%EB%A0%A5>

214)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%ED%95%B4%EC%96%91%EA%B5%90%ED%86%B5%EC%95%88%EC%A0%84%EA%B3%B5%EB%8B%A8>

215)
<https://ko.wikipedia.org/wiki/%ED%95%9C%EA%B5%AD%EC%A7%80%EC%97%AD%EB%82%9C%EB%B0%A9%EA%B3%B5%EC%82%A>
C

216) <https://www.komeri.re.kr/content.php?id=109030>

217)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%ED%95%B4%EC%96%91%EA%B3%BC%ED%95%99%EA%B8%B0%EC%88%A0%EC%9B%90>

218)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%A7%80%EC%A7%88%EC%9E%90%EC%9B%90%EC%97%B0%EA%B5%AC%EC%9B%90>

References

- 219)
<https://namu.wiki/w/%EA%B5%AD%EB%A6%BD%ED%95%9C%EA%B5%AD%ED%95%B4%EC%96%91%EB%8C%80%ED%95%99%EA%B5%90>
- 220) <https://namu.wiki/w/%EB%8C%80%ED%95%9C%EB%AF%BC%EA%B5%AD%20%EA%B8%B0%EC%83%81%EC%B2%AD>
- 221) https://ko.wikipedia.org/wiki/%ED%8C%8C%EC%9D%BC:ADD_Logo_%28CI%29.svg
- 222)
https://ko.wikipedia.org/wiki/%EB%8C%80%ED%95%9C%EB%AF%BC%EA%B5%AD_%EA%B5%AD%EA%B0%80%EC%A0%95%EB%B3%B4%EC%9B%90
- 223) <https://namu.wiki/w/%EA%B5%AD%EB%B0%A9%EA%B3%BC%ED%95%99%EC%97%B0%EA%B5%AC%EC%86%8C>
- 224) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EA%B0%80%EC%8A%A4%EC%95%88%EC%A0%84%EA%B3%B5%EC%82%AC>
- 225) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EA%B0%80%EC%8A%A4%EA%B8%B0%EC%88%A0%EA%B3%B5%EC%82%AC>
- 226)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%ED%95%B4%EC%96%91%EC%88%98%EC%82%B0%EC%97%B0%EC%88%98%EC%9B%90>

References

227) <https://namu.wiki/w/%ED%95%B4%EC%96%91%ED%99%98%EA%B2%BD%EA%B3%B5%EB%8B%A8>

228)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%ED%95%B4%EC%96%91%EC%88%98%EC%82%B0%EA%B0%9C%EB%B0%9C%EC%9B%90>

229) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EA%B8%B0%EA%B3%84%EC%97%B0%EA%B5%AC%EC%9B%90>

230)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%97%90%EB%84%88%EC%A7%80%EA%B8%B0%EC%88%A0%EC%97%B0%EA%B5%AC%EC%9B%90>

231) <https://jaseol.com/companies/15049>

232)
<https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%A0%84%EC%9E%90%ED%86%B5%EC%8B%A0%EC%97%B0%EA%B5%AC%EC%9B%90>

233) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%A0%84%EA%B8%B0%EC%97%B0%EA%B5%AC%EC%9B%90>

234) <http://www.komea.kr/koema/BbsPopup.do?bCode=B011&idx=36847&block=36847&rlevel=0&boss=0>

References

- 235) <https://namu.wiki/w/%ED%95%9C%EA%B5%AD%EC%83%9D%EC%82%B0%EA%B8%B0%EC%88%A0%EC%97%B0%EA%B5%AC%EC%9B%90>
- 236) https://en.wikipedia.org/wiki/Texas_A%26M_University
- 237) https://fr.wikipedia.org/wiki/Universit%C3%A9_de_Strathclyde
- 238) https://en.wikipedia.org/wiki/Florida_Institute_of_Technology
- 239) <https://logos-download.com/77747-nanyang-technological-university-logo-download.html>
- 240) <https://commons.wikimedia.org/wiki/File:NTNU-logo.svg>
- 241) https://en.m.wikipedia.org/wiki/File:Virginia_Tech_logo.svg
- 242) <https://www.eescable.com/product/twisted-flexible-electric-wire/?srsId=AfmBOopU9t3L6OZpnN8zAMB6NEFxLI6kxGLdSLWGz0-cBlOeJv1Agp0X>
- 243) <https://swacable.net/1-50-2-5mm-x-6-5mm-blue-ring-terminal-cable-lugs.html>



References

- 244) <https://ko.aliexpress.com/item/1005007188684964.html?gatewayAdapt=glo2kor>
- 245) <https://www.amazon.de/-/en/Insulated-Electrical-Connection-Connector-Connectors/dp/B0CHVBX68V?th=1>
- 246) <https://www.bearingstation.co.uk/products/internal-circlip-1300-standard-steel-19mm-shaft=29876>
- 247) <https://kr.rs-online.com/web/p/pressure-gauges/2228576>
- 248) <https://www.bimetalgauge.com/showroom/v-line-glass-marine-thermometer.html>
- 249) <https://www.youtube.com/watch?v=X1elipmyaWc>
- 250) <https://www.theengineeringchoice.com/how-to-read-vernier-caliper/>
- 251) <https://redfluid.es/en/how-measure-with-caliper/>



References

- 252) <https://www.webassign.net/labsgraceperiod/ncsulcpmech2/appendices/appendixD/appendixD.html>
- 253) <https://mechlesson.com/vernier-caliper/>
- 254) <https://alandia.com/article/safe-anchoring/>
- 255) <https://maritimesa.org/nautical-science-grade-11/2020/10/27/anchoring/>
- 256) https://www.dh.co.kr/about_us/overview.php
- 257) https://www.worldpanasia.com/about/about_ci.php
- 258) <https://www.motorship.com/wingd/1074.bio>
- 259) <https://www.brandb.net/archive/everllence-2025-06>
- 260) <https://ko.wikipedia.org/wiki/%EB%B0%94%EB%A5%B4%EC%A7%88%EB%9D%BC>



References

261) <https://www.youtube.com/channel/UCIv6et9zciX0LD-72qKboFg/about>

262) <https://namu.wiki/w/%EA%B5%AD%EB%A6%BD%EA%B5%B0%EC%82%B0%EB%8C%80%ED%95%99%EA%B5%90>

263) <https://namu.wiki/w/%EC%A0%84%EB%82%A8%EB%8C%80%ED%95%99%EA%B5%90>

264) <https://namu.wiki/w/%EA%B5%AD%EB%A6%BD%EB%B6%80%EA%B2%BD%EB%8C%80%ED%95%99%EA%B5%90>

265) https://hijob.pen.go.kr/dajobgo/2023/01_school_view5.asp?school=5

266) <https://namu.wiki/w/%EC%A0%9C%EC%A3%BC%EB%8C%80%ED%95%99%EA%B5%90>

267) <https://www.facebook.com/inm.icehs.kr/>



References

268)

<https://namu.wiki/w/%EA%B2%BD%EC%83%81%EA%B5%AD%EB%A6%BD%EB%8C%80%ED%95%99%EA%B5%90>

269) https://www.instagram.com/_haemago/

270)

<https://namu.wiki/w/%EC%9D%B8%EC%B2%9C%ED%95%B4%EC%96%91%EA%B3%BC%ED%95%99%EA%B3%A0%EB%93%B1%ED%95%99%EA%B5%90>

271)

<https://www.facebook.com/p/%EC%B6%A9%EB%82%A8%ED%95%B4%EC%96%91%EA%B3%BC%ED%95%99%EA%B3%A0%EB%93%B1%ED%95%99%EA%B5%90-100054616503396/>

272) <https://blog.naver.com/newtnt77/222964761596>

273) <https://blog.naver.com/newtnt77/222770197357>



References

- 274) <https://m.blog.naver.com/newtnt77/222859123508>
- 275) <https://m.blog.naver.com/newtnt77/222859584663>
- 276) <https://www.amp.gob.pa/home-fred/attachment/panama-ship-registry/>
- 277) <https://www.slide2open.net/en/sponsor/iri-the-marshall-islands-registry/>
- 278) <https://www.liscr.com/>
- 279) https://m.saramin.co.kr/job-search/company-info-view/csn/Rkw0SzRRZ0V6ejJxL3htYUtYenZKUT09/company_nm/%28%EC%9E%AC%29%ED%95%9C%EA%B5%AD%ED%95%B4%EC%82%AC%ED%98%91%EB%A0%A5%EC%84%BC%ED%84%B0
- 280) <https://www.jonghap.co.kr/new/01intro/02.asp>
- 281) <https://www.joongang.co.kr/article/23873698>



References

- 282) <https://www.facebook.com/Lab021Vessellink/>
- 283) https://kgaofflinecenter.com/dt_logos/rolls-royce-marine/
- 284) https://kgaofflinecenter.com/dt_logos/rolls-royce-marine/
- 285) <https://www.energydaily.co.kr/news/articleView.html?idxno=126231>
- 286) https://ko.wikipedia.org/wiki/ABB_%EA%B7%B8%EB%A3%B9
- 287) <https://www.kcg.go.kr/kcg/cm/cntnts/cntntsView.do?mi=2825&cntntsId=63>
- 288) https://koreaenergyshow.energy.or.kr/program/participate_view.do?no=8283
- 289) <https://namu.wiki/w/%EC%84%9C%EC%9A%B8%EB%8C%80%ED%95%99%EA%B5%90>
- 290) <https://ko.wikipedia.org/wiki/%EB%B6%80%EC%82%B0%EB%8C%80%ED%95%99%EA%B5%90>
- 291) <https://namu.wiki/w/%EC%9D%B8%ED%95%98%EB%8C%80%ED%95%99%EA%B5%90>
- 292) <https://www.boyiprototyping.com/blogs/what-is-a-flange/>

